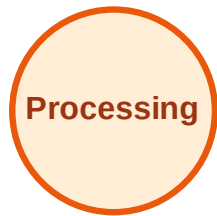


BFS Traversal

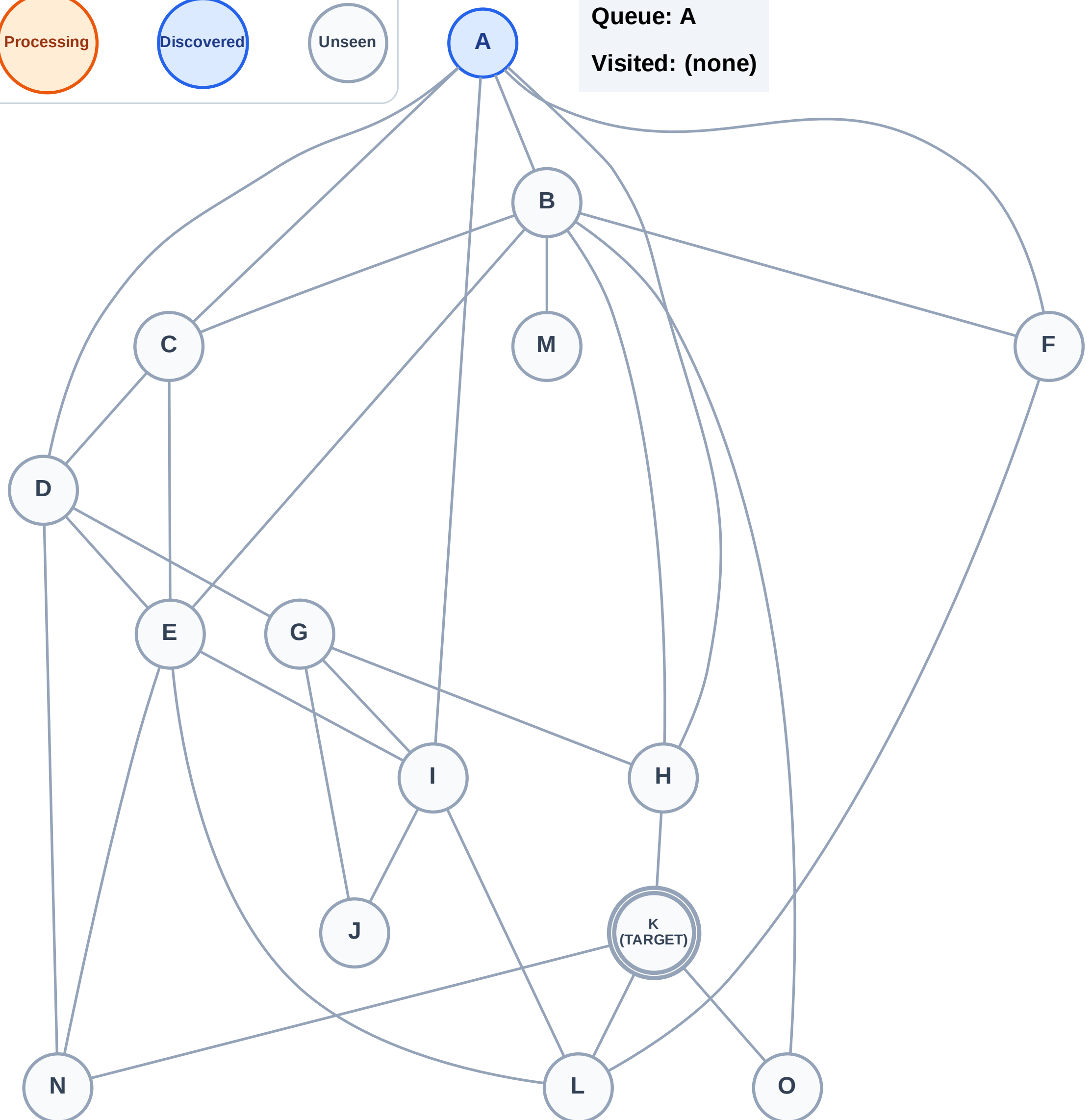
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

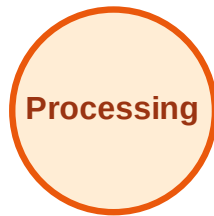
Visited: (none)



BFS Traversal

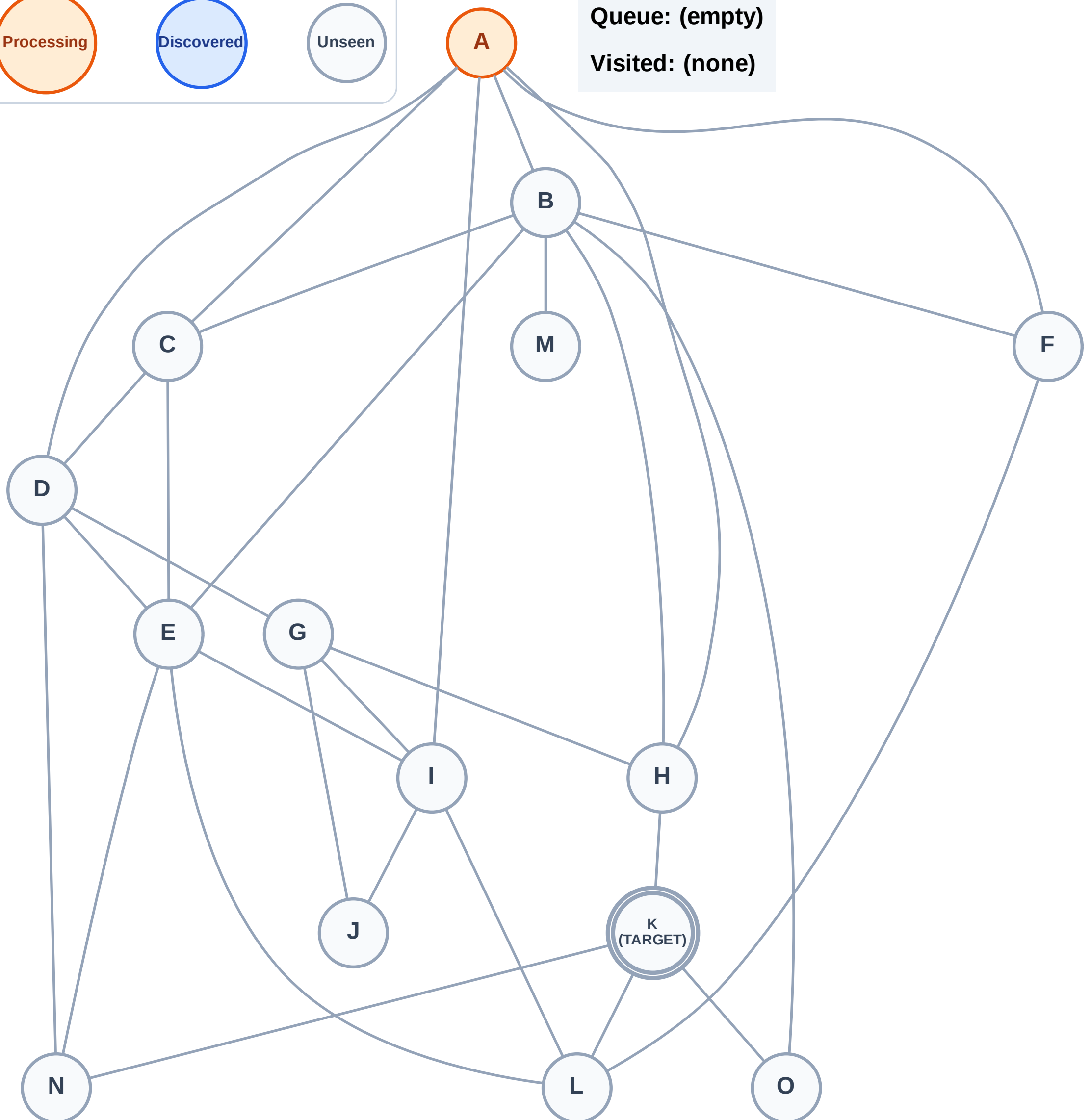
Step 2: Process node A

Legend



Queue: (empty)

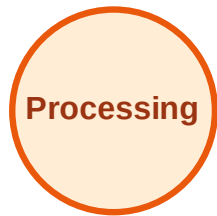
Visited: (none)



BFS Traversal

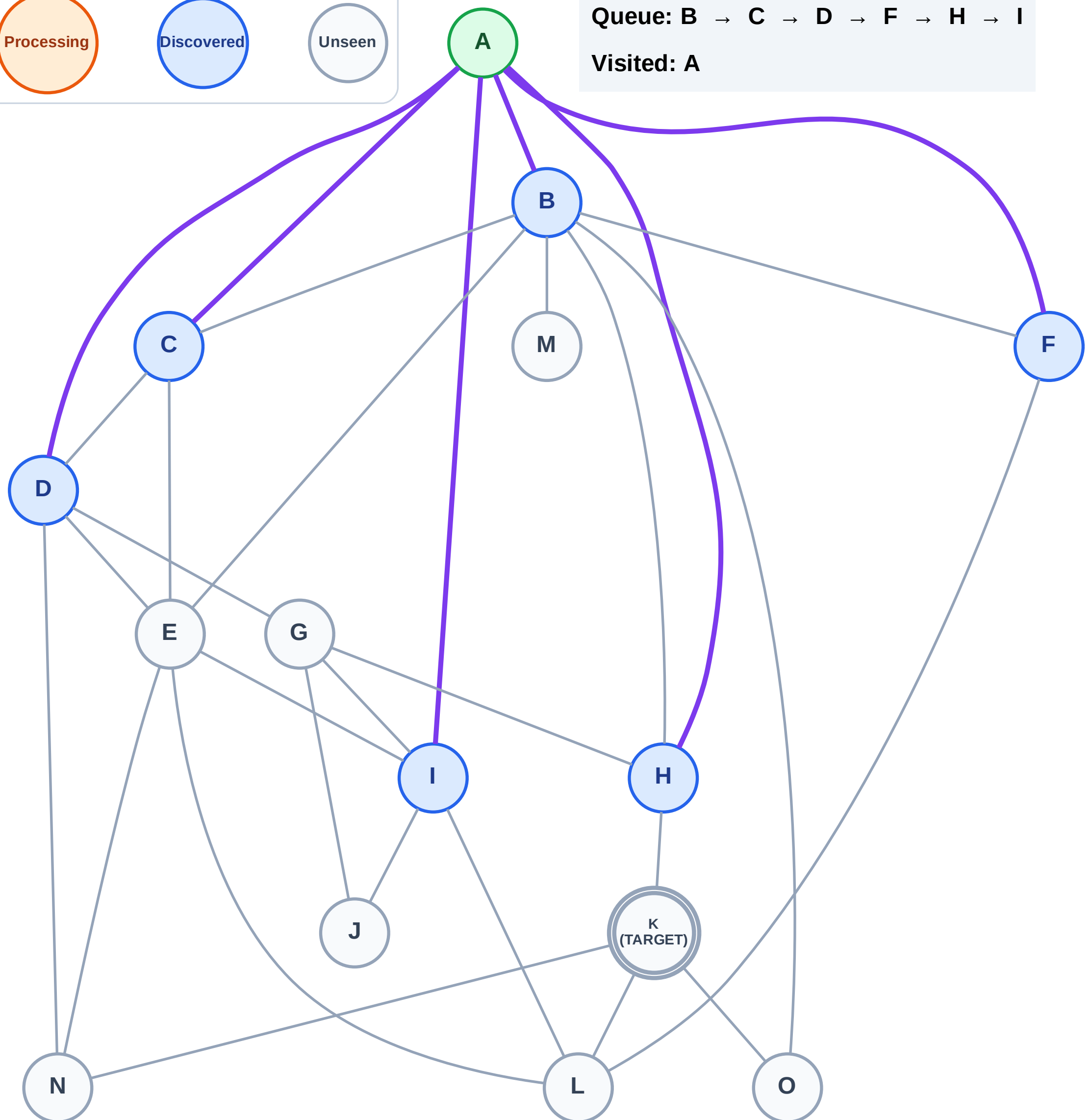
Step 3: Discover B, C, D, F, H, I from A

Legend



Queue: B → C → D → F → H → I

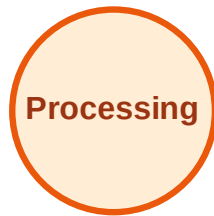
Visited: A



BFS Traversal

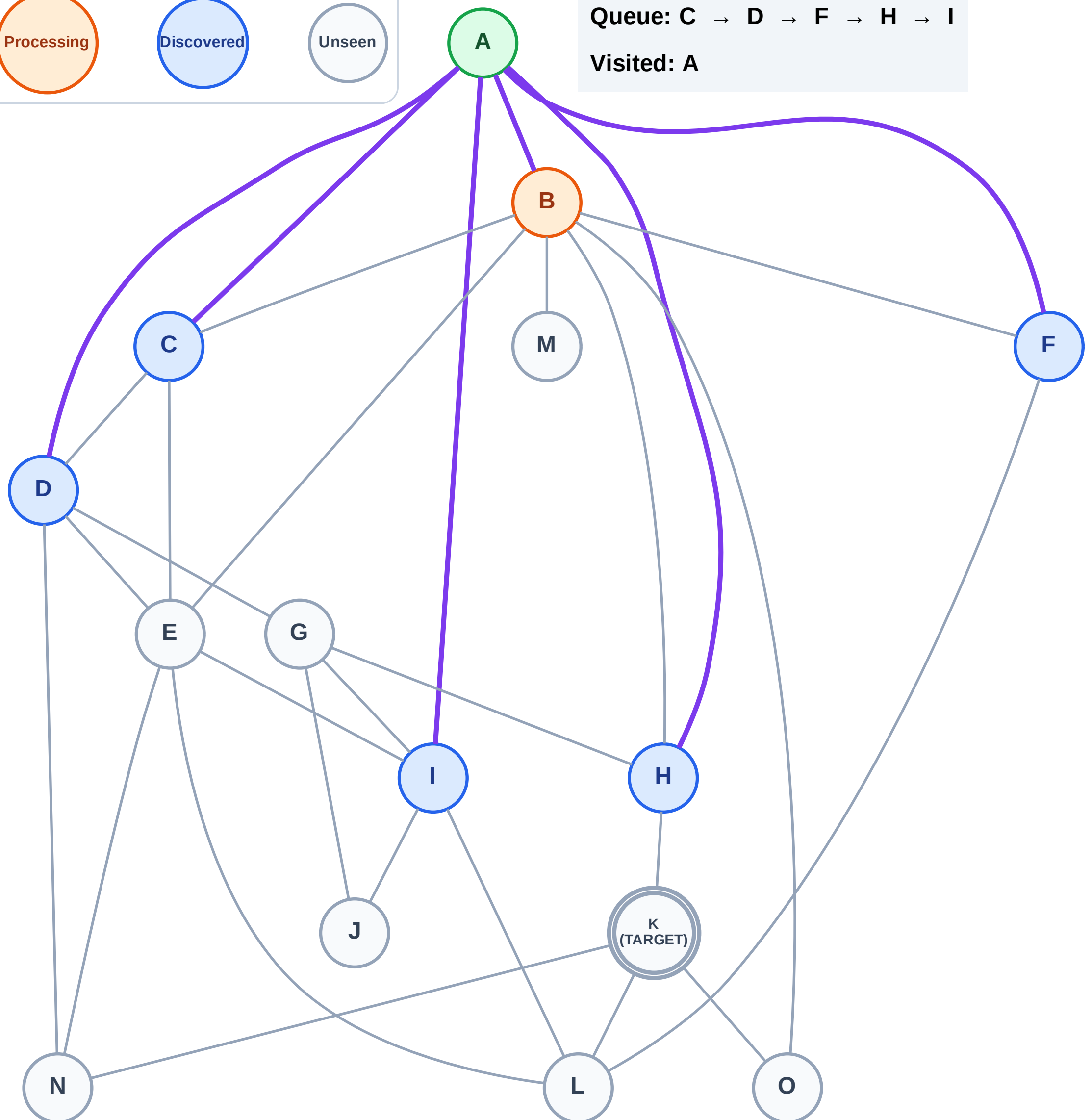
Step 4: Process node B

Legend



Queue: C → D → F → H → I

Visited: A



BFS Traversal

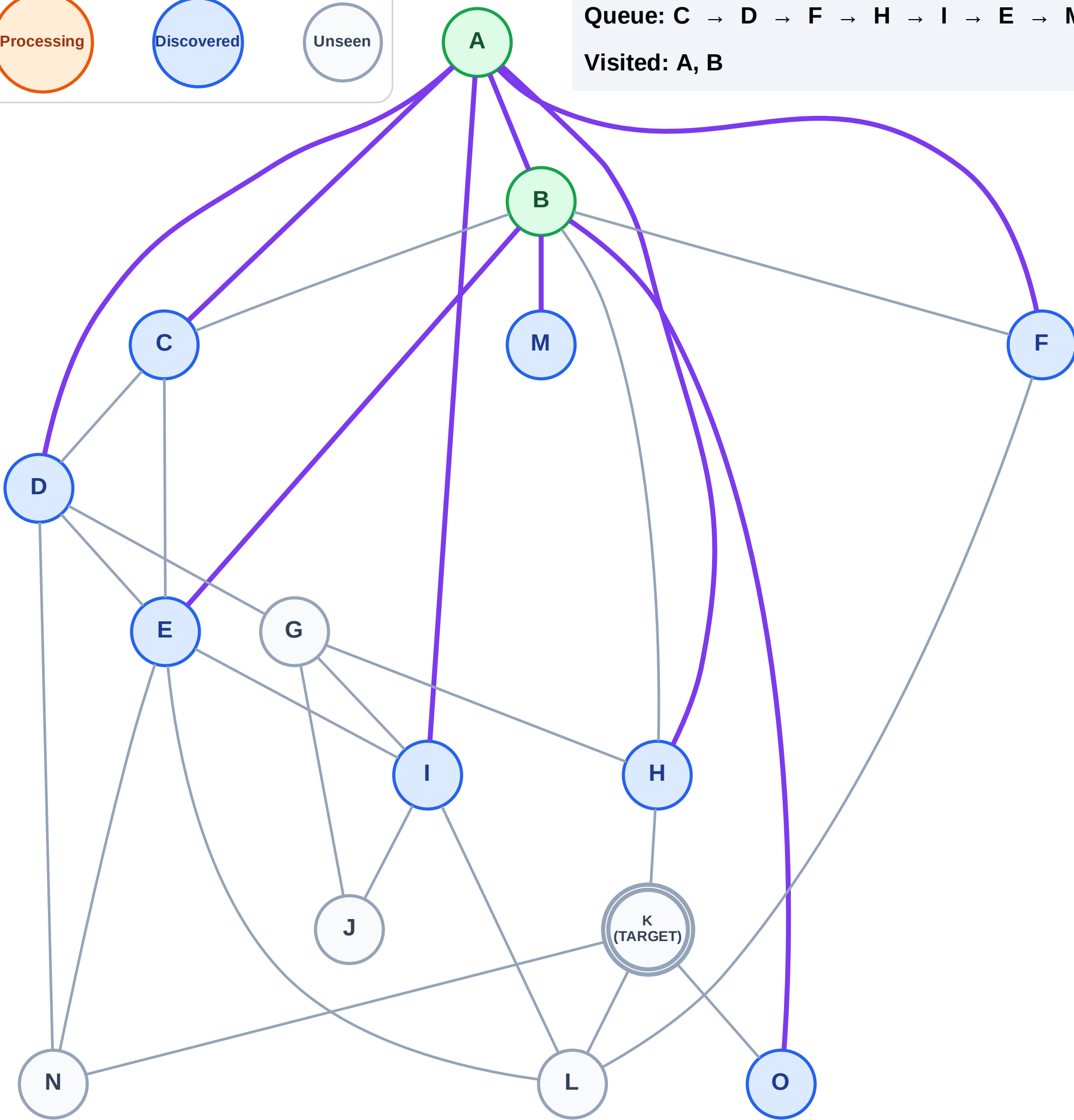
Step 5: Discover E, M, O from B

Legend



Queue: C → D → F → H → I → E → M → O

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

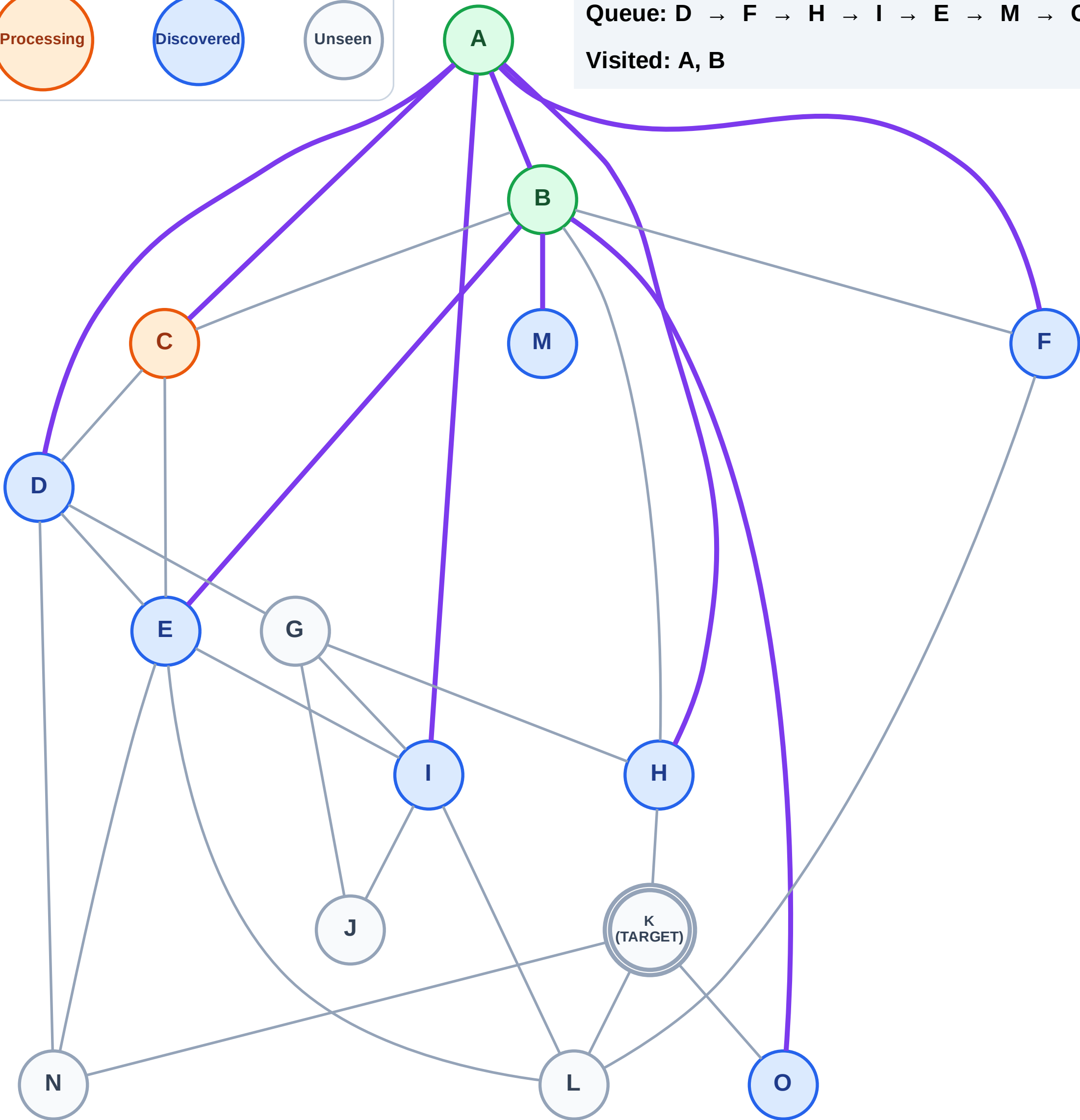
Processing

Discovered

Unseen

Queue: D → F → H → I → E → M → O

Visited: A, B



BFS Traversal

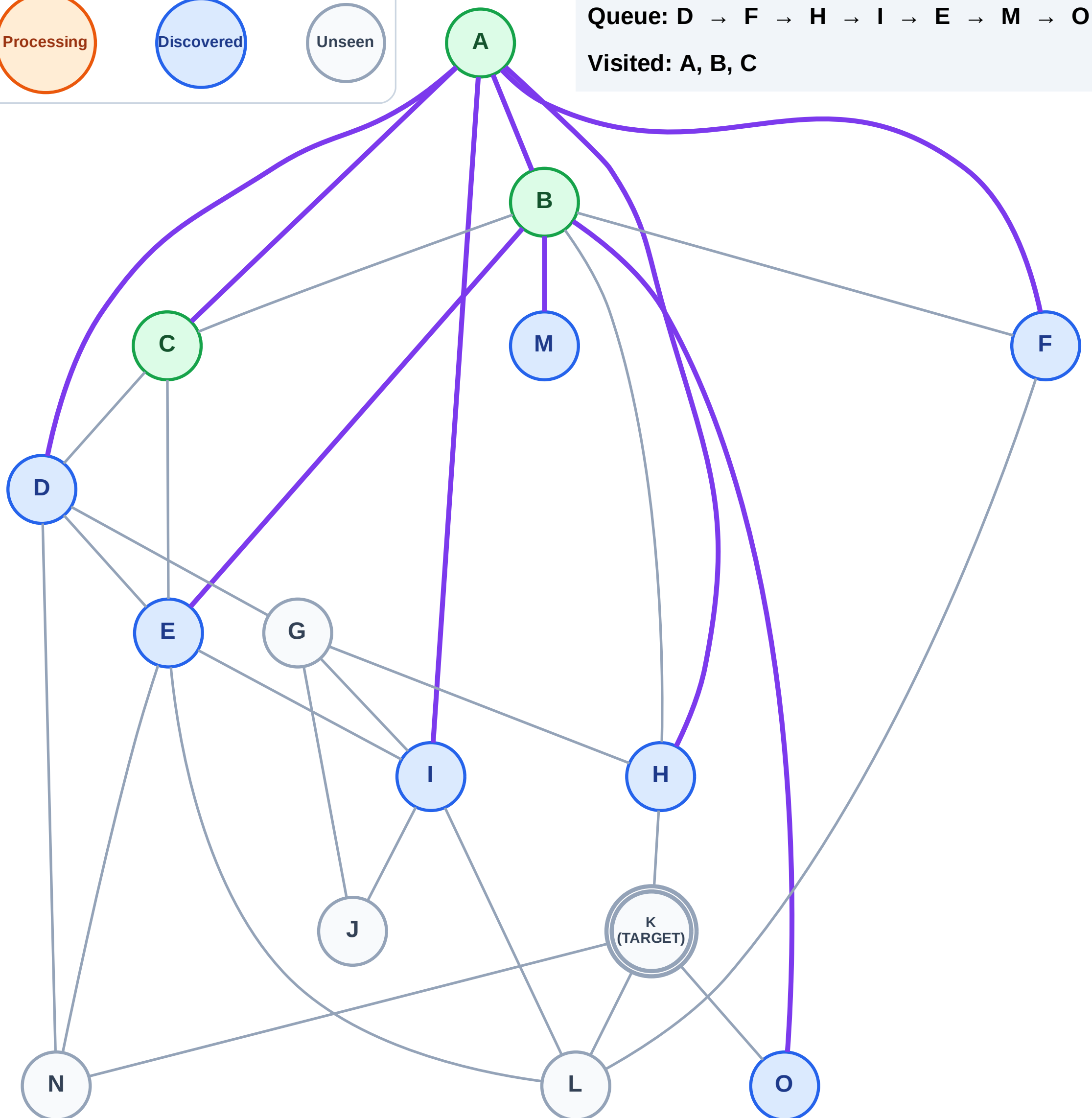
Step 7: No new neighbors from C

Legend



Queue: D → F → H → I → E → M → O

Visited: A, B, C



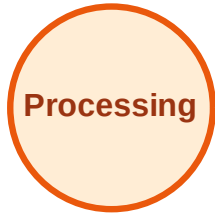
BFS Traversal

Step 8: Process node D

Legend



Complete



Processing



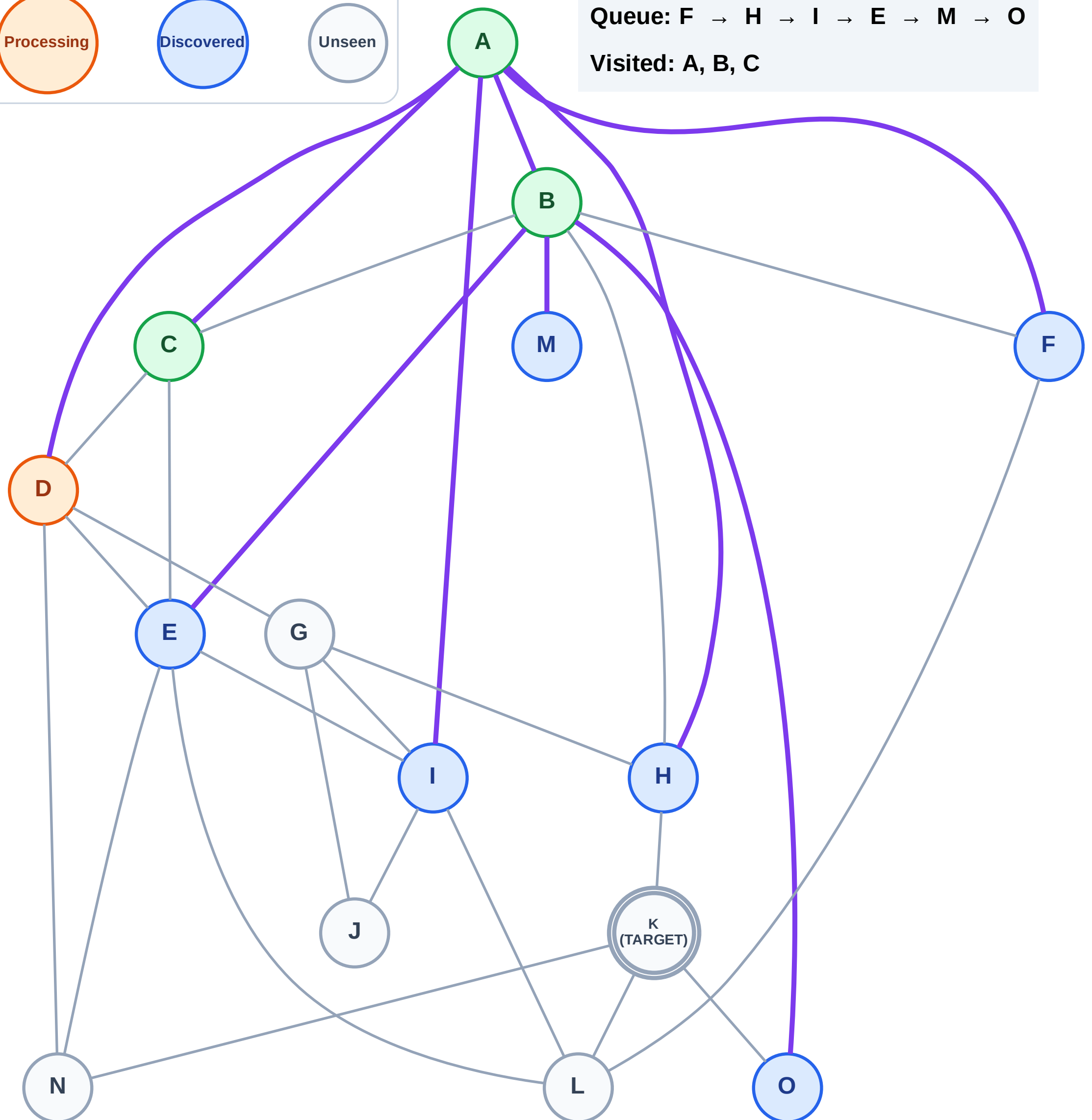
Discovered



Unseen

Queue: F → H → I → E → M → O

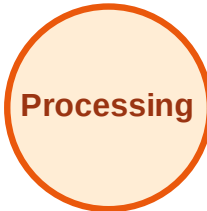
Visited: A, B, C



BFS Traversal

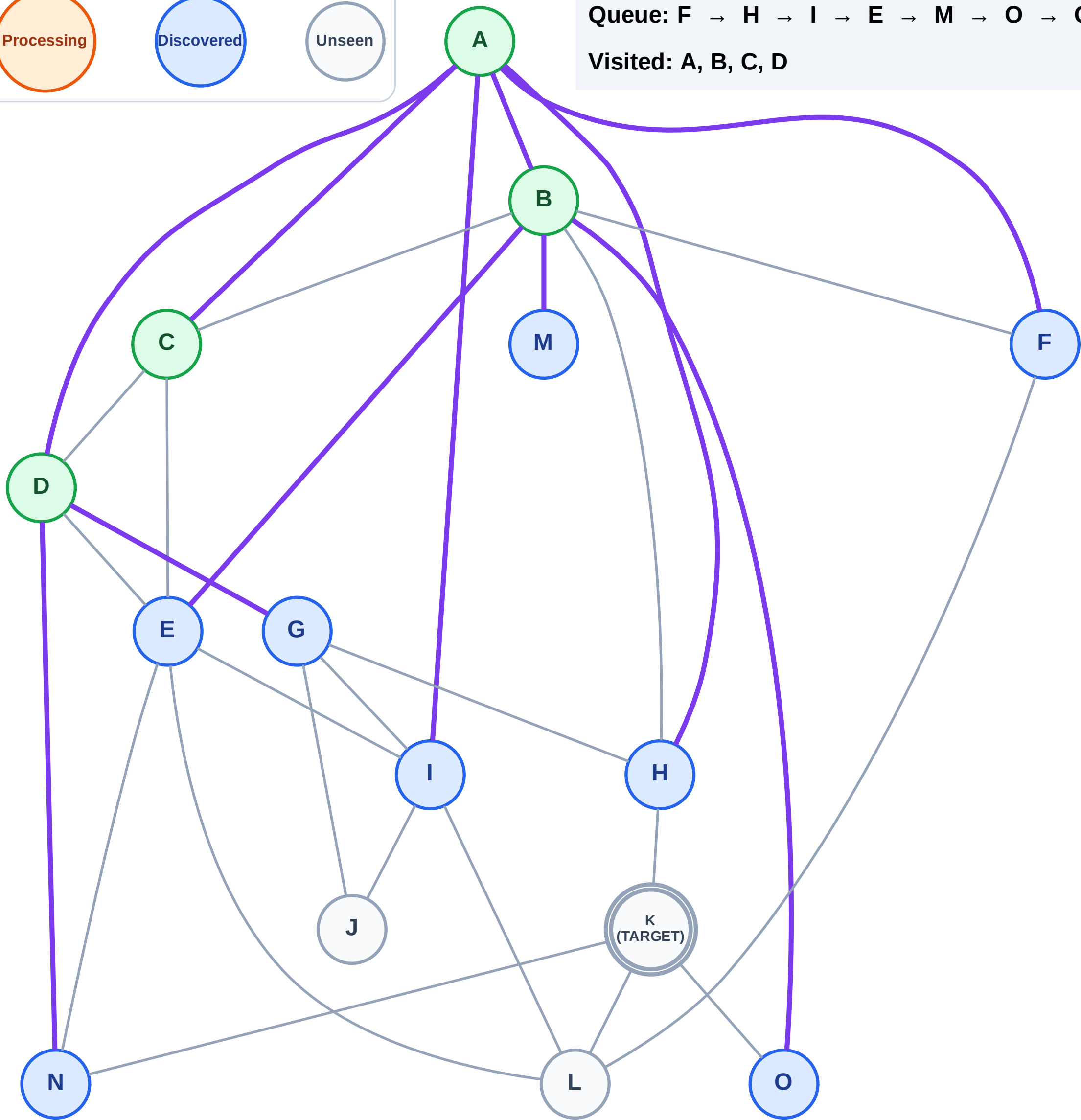
Step 9: Discover G, N from D

Legend



Queue: F → H → I → E → M → O → G → N

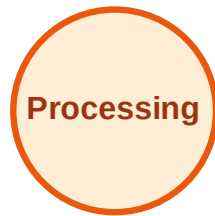
Visited: A, B, C, D



BFS Traversal

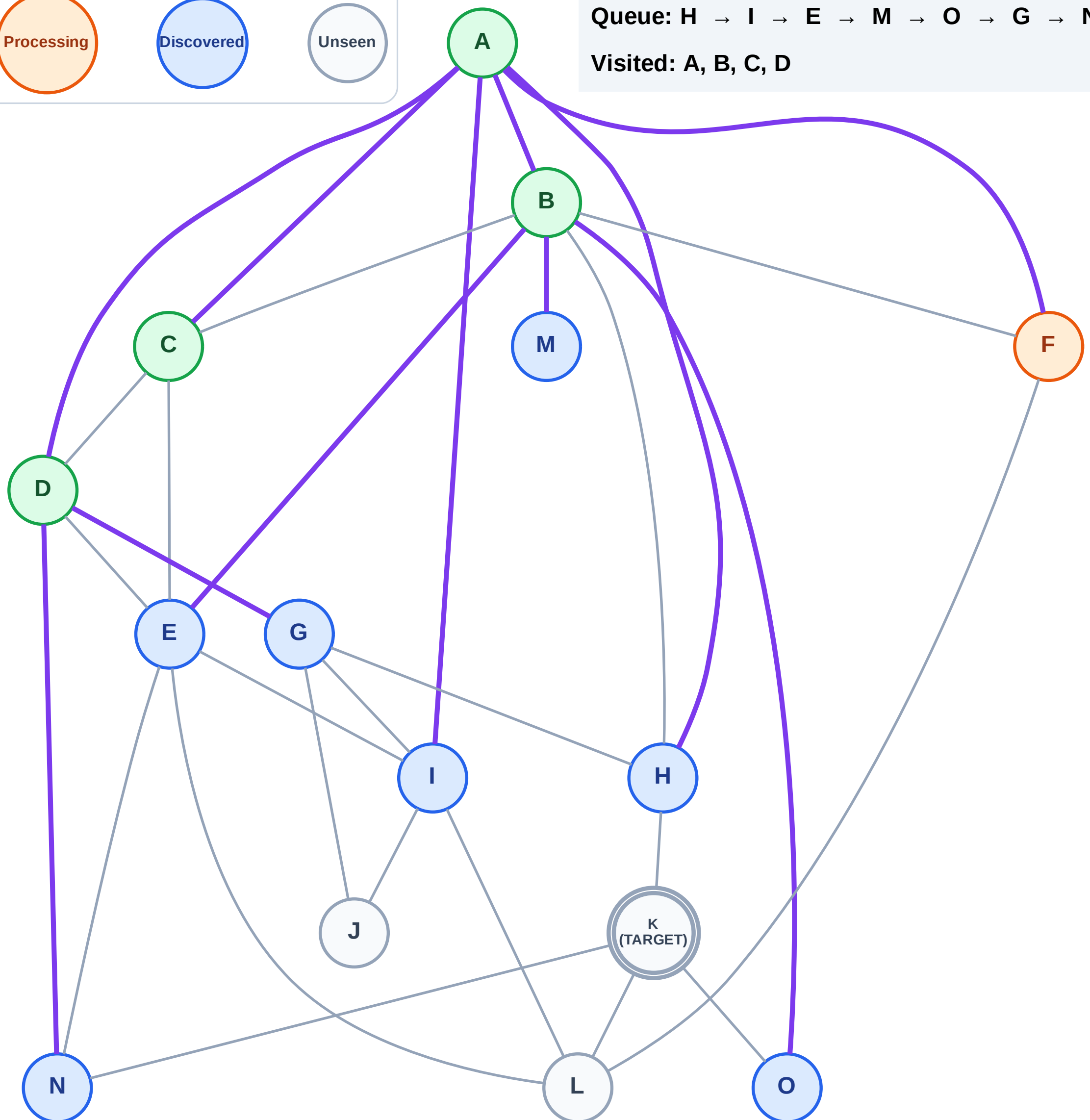
Step 10: Process node F

Legend



Queue: H → I → E → M → O → G → N

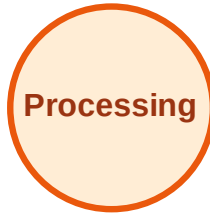
Visited: A, B, C, D



BFS Traversal

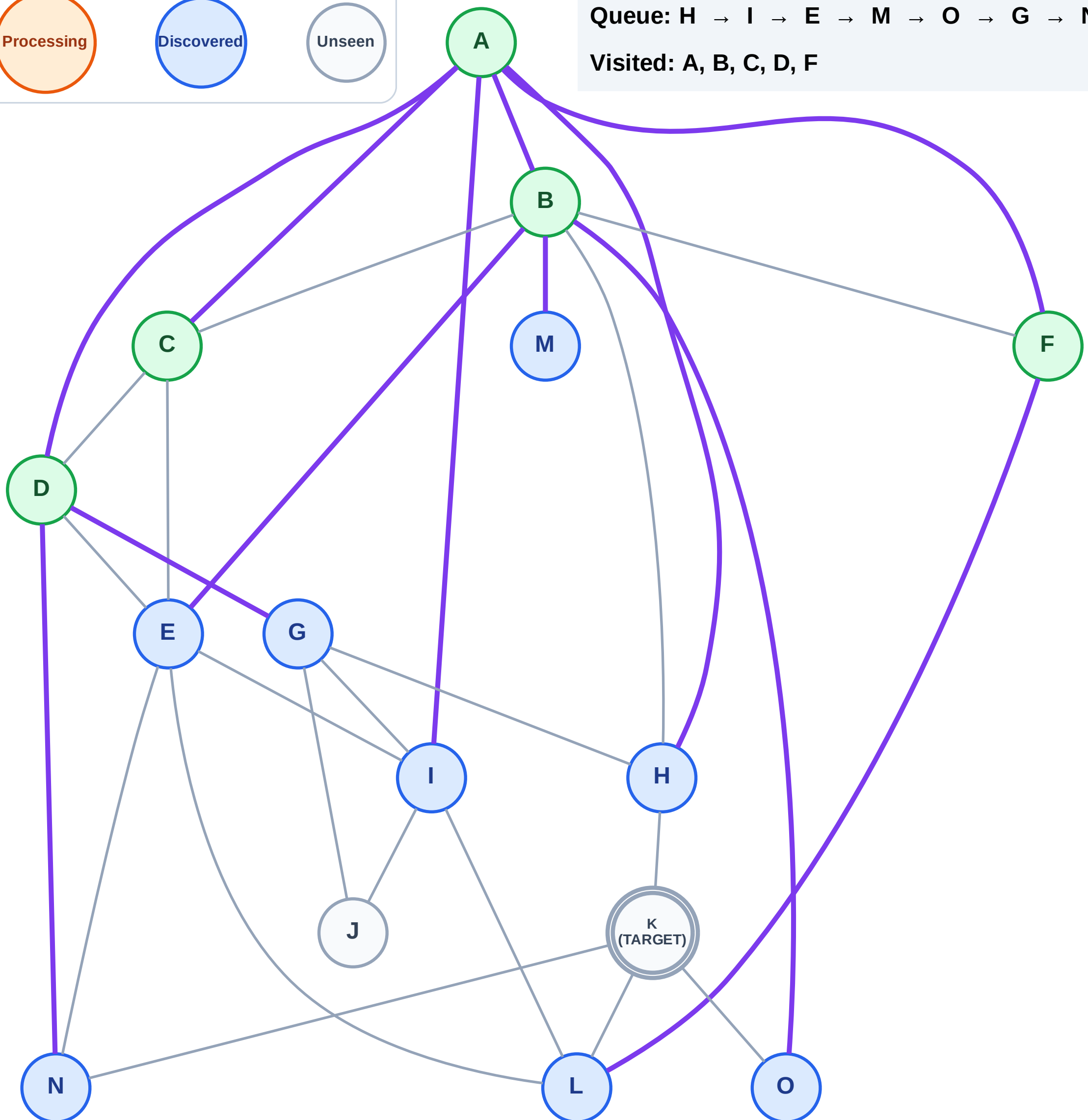
Step 11: Discover L from F

Legend



Queue: H → I → E → M → O → G → N → L

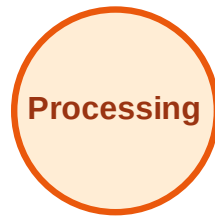
Visited: A, B, C, D, F



BFS Traversal

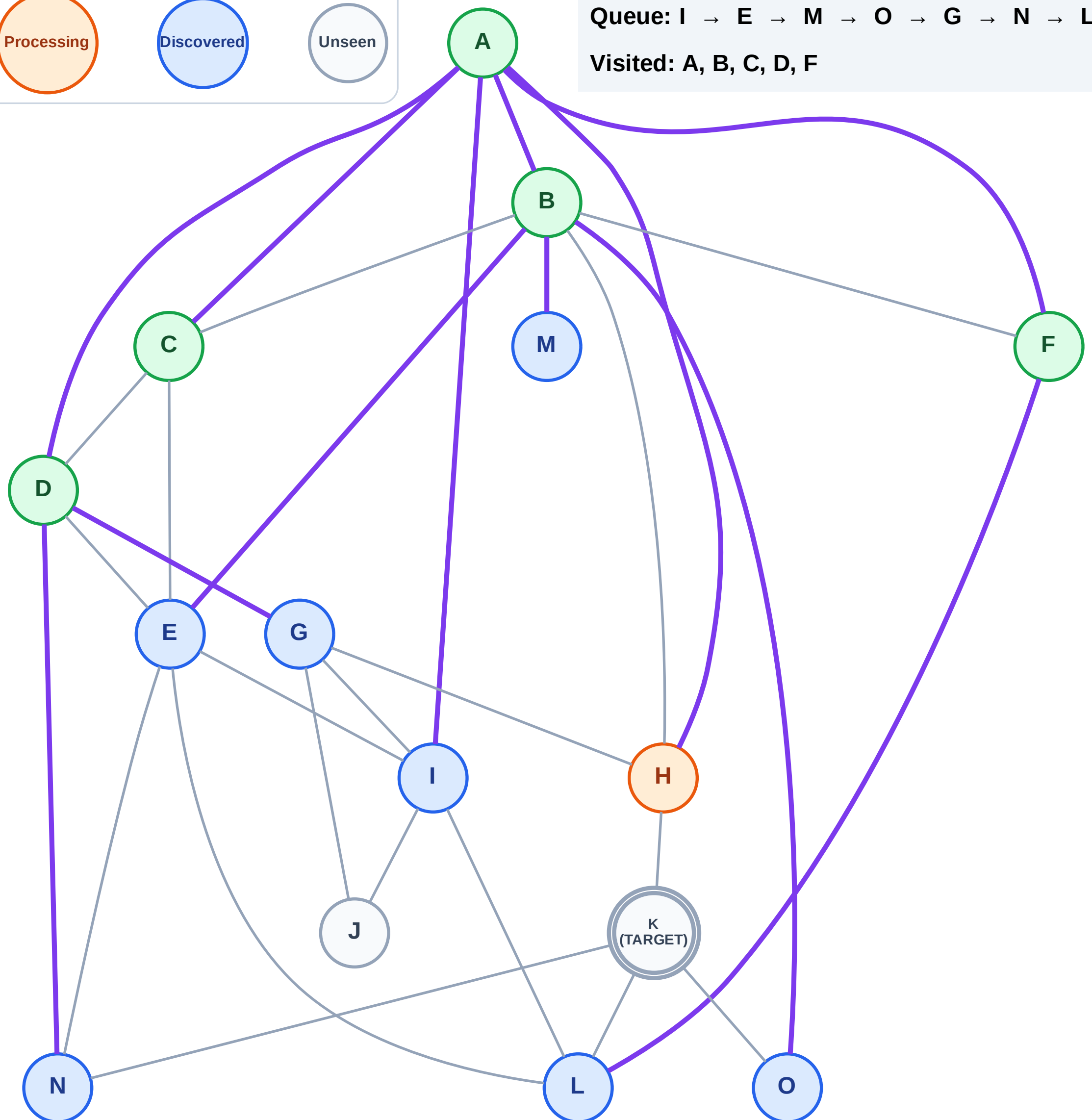
Step 12: Process node H

Legend



Queue: I → E → M → O → G → N → L

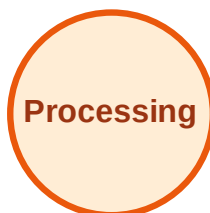
Visited: A, B, C, D, F



BFS Traversal

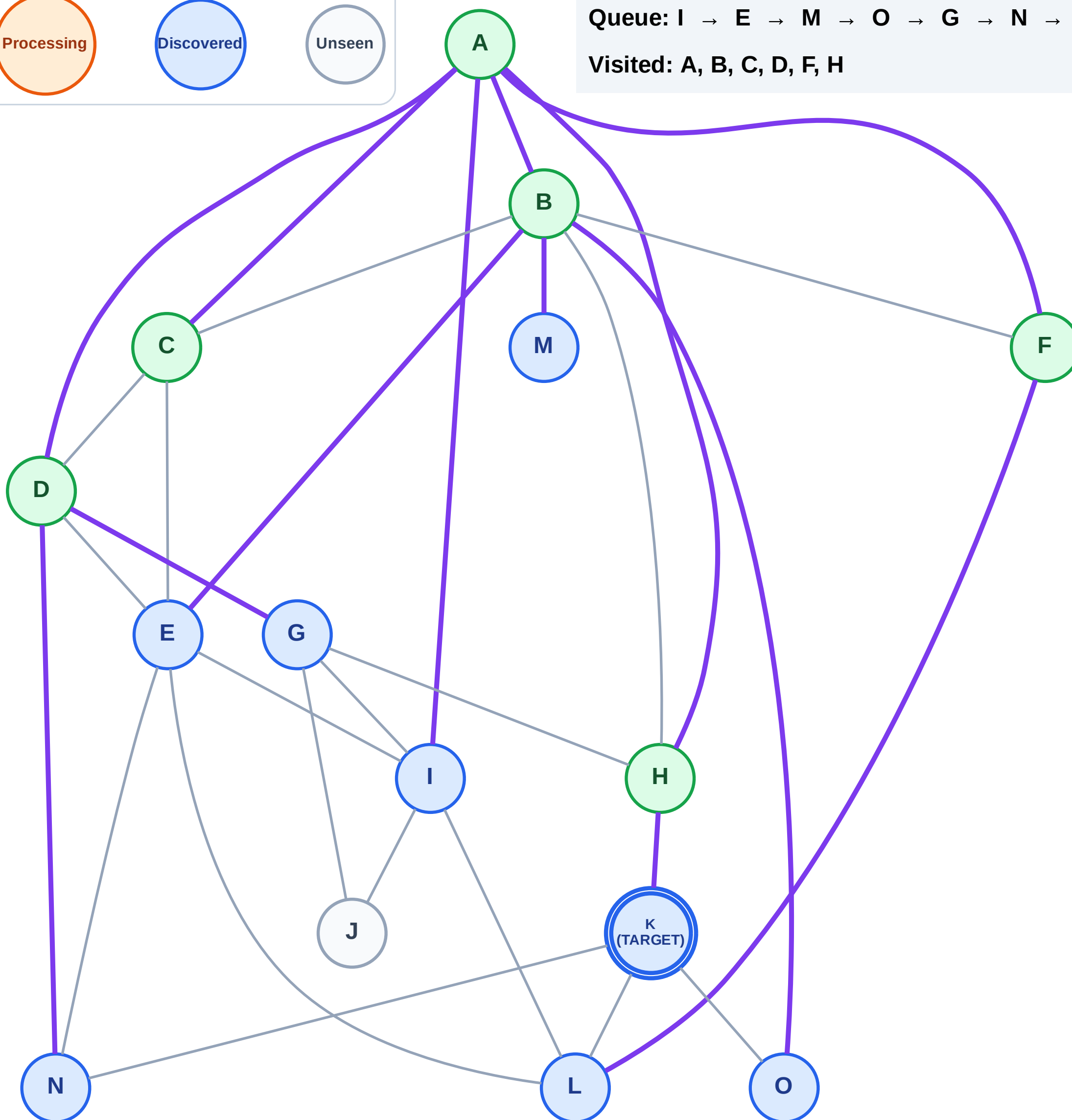
Step 13: Discover K from H

Legend



Queue: I → E → M → O → G → N → L → K

Visited: A, B, C, D, F, H



BFS Traversal

Step 14: Process node I

Legend

Complete

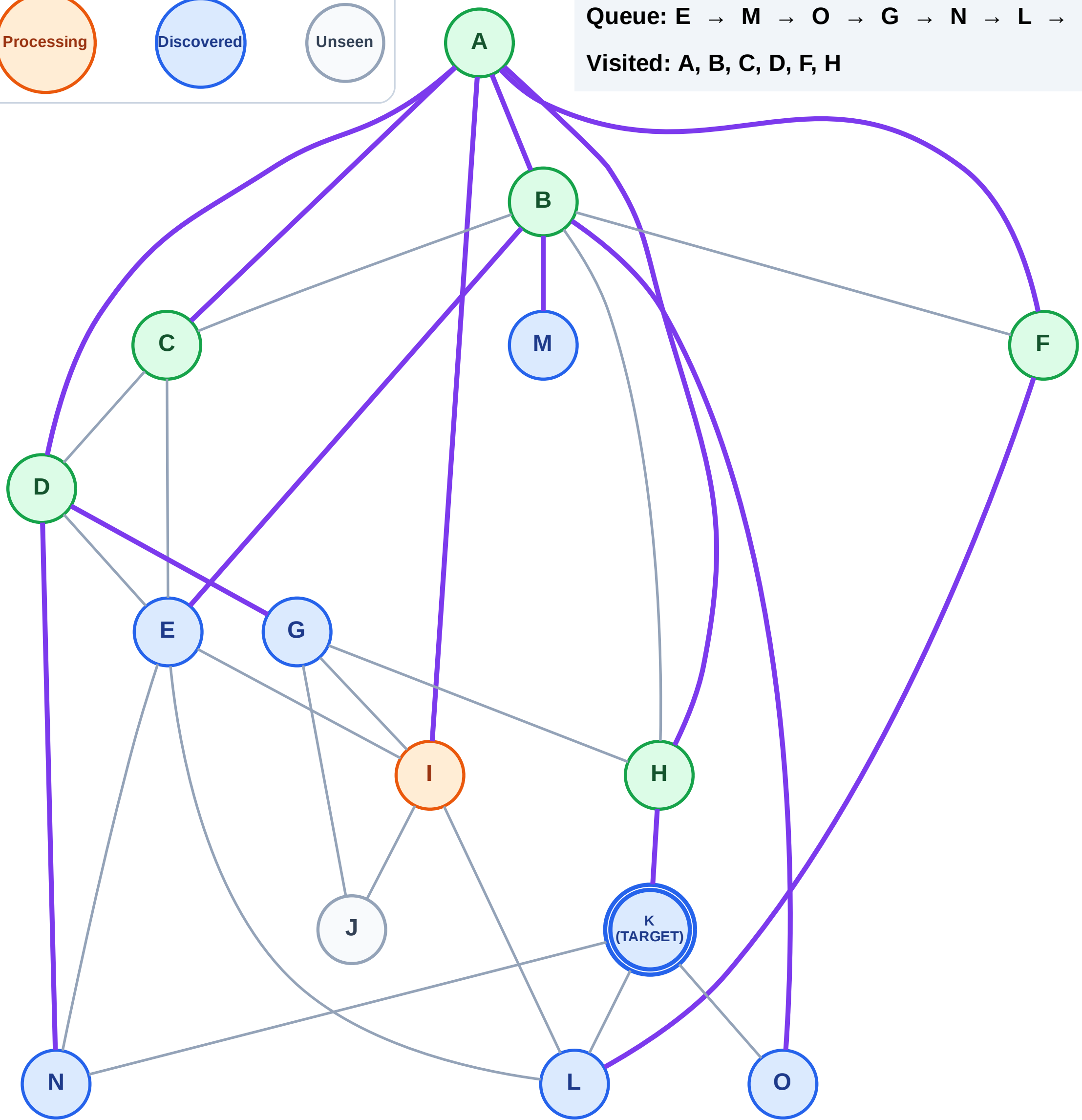
Processing

Discovered

Unseen

Queue: E → M → O → G → N → L → K

Visited: A, B, C, D, F, H



BFS Traversal

Step 15: Discover J from I

Legend



Complete



Processing



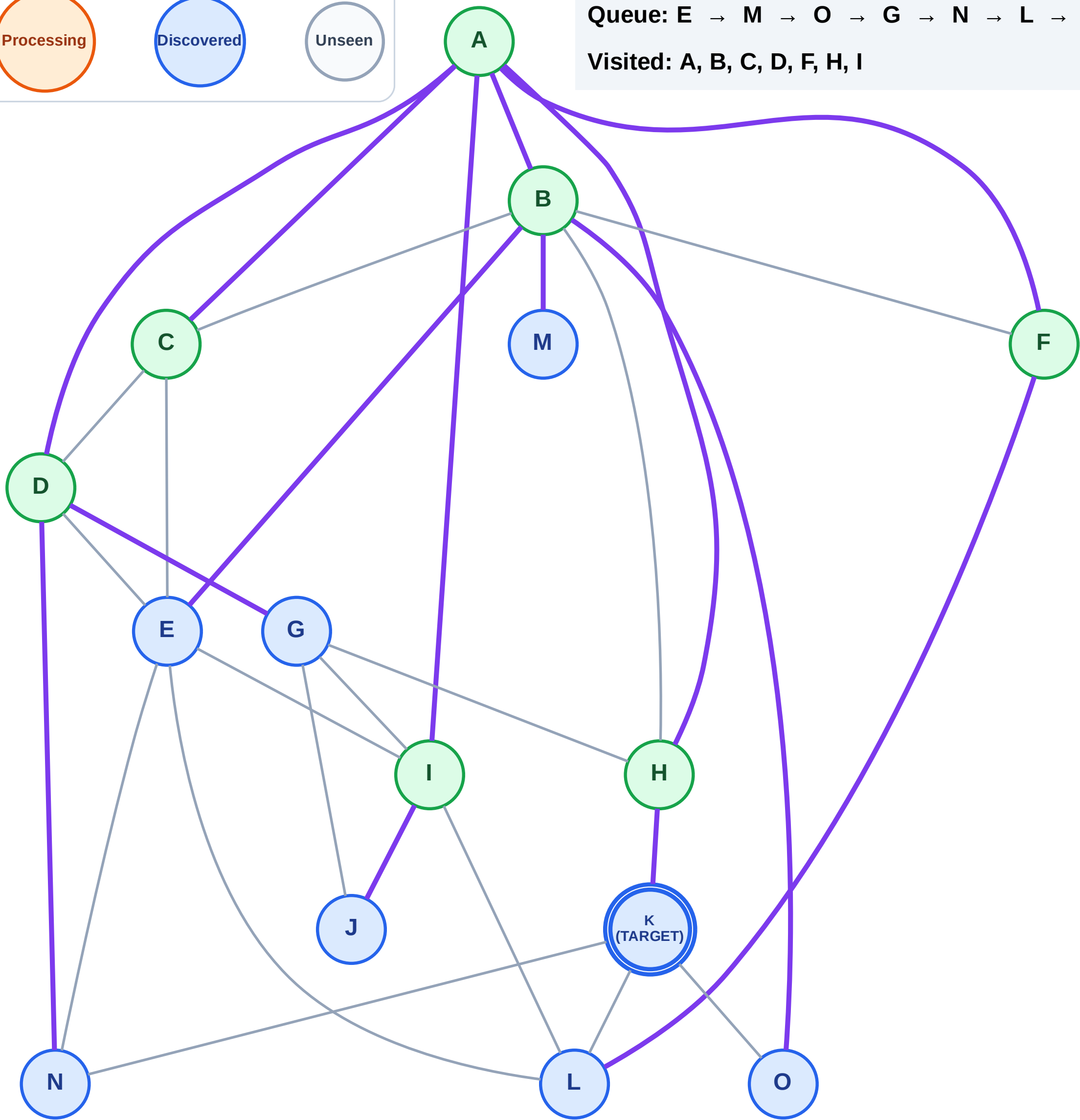
Discovered



Unseen

Queue: E → M → O → G → N → L → K → J

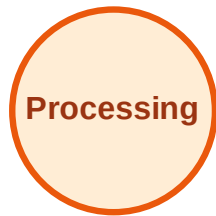
Visited: A, B, C, D, F, H, I



BFS Traversal

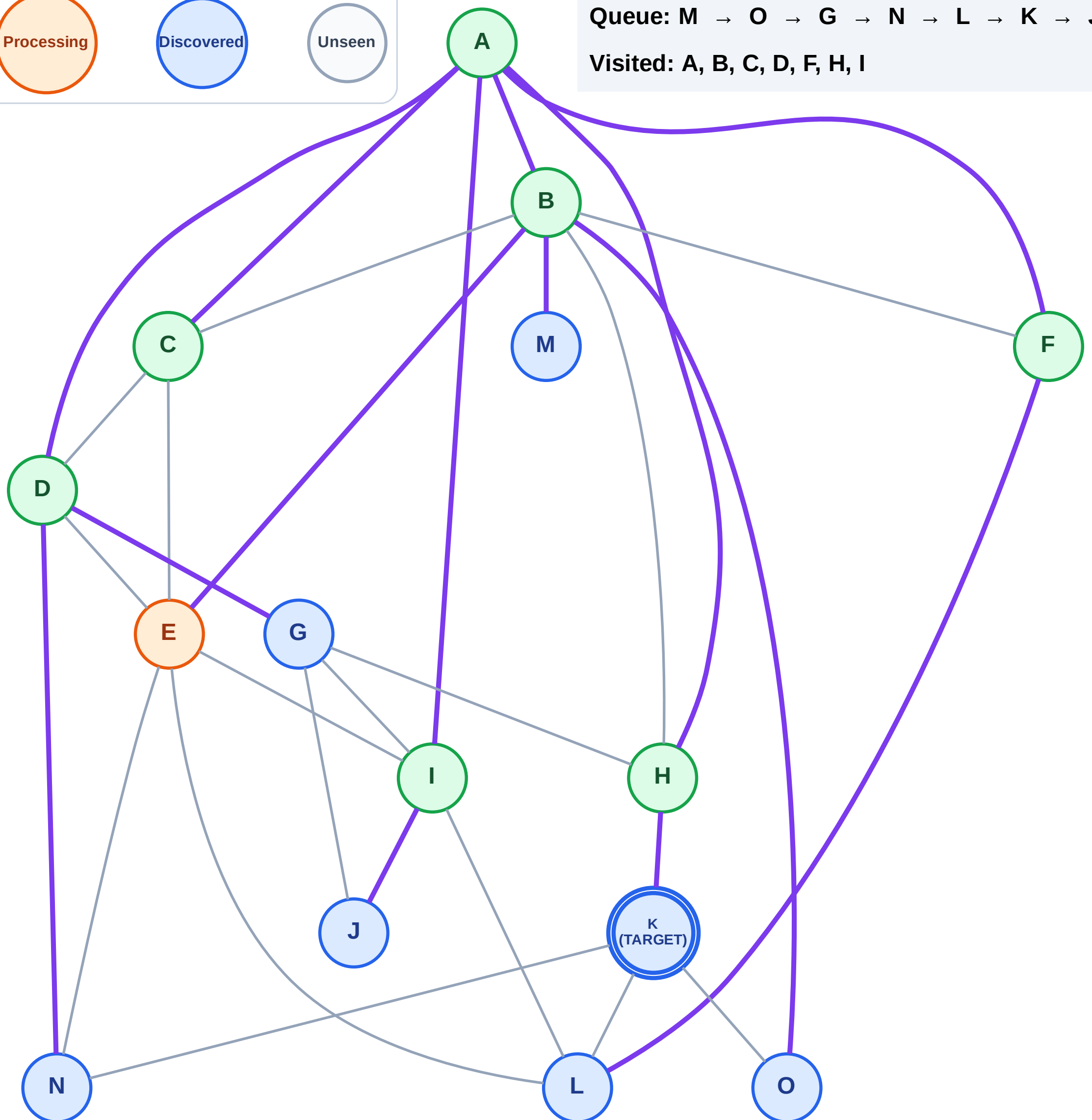
Step 16: Process node E

Legend



Queue: M → O → G → N → L → K → J

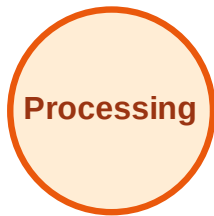
Visited: A, B, C, D, F, H, I



BFS Traversal

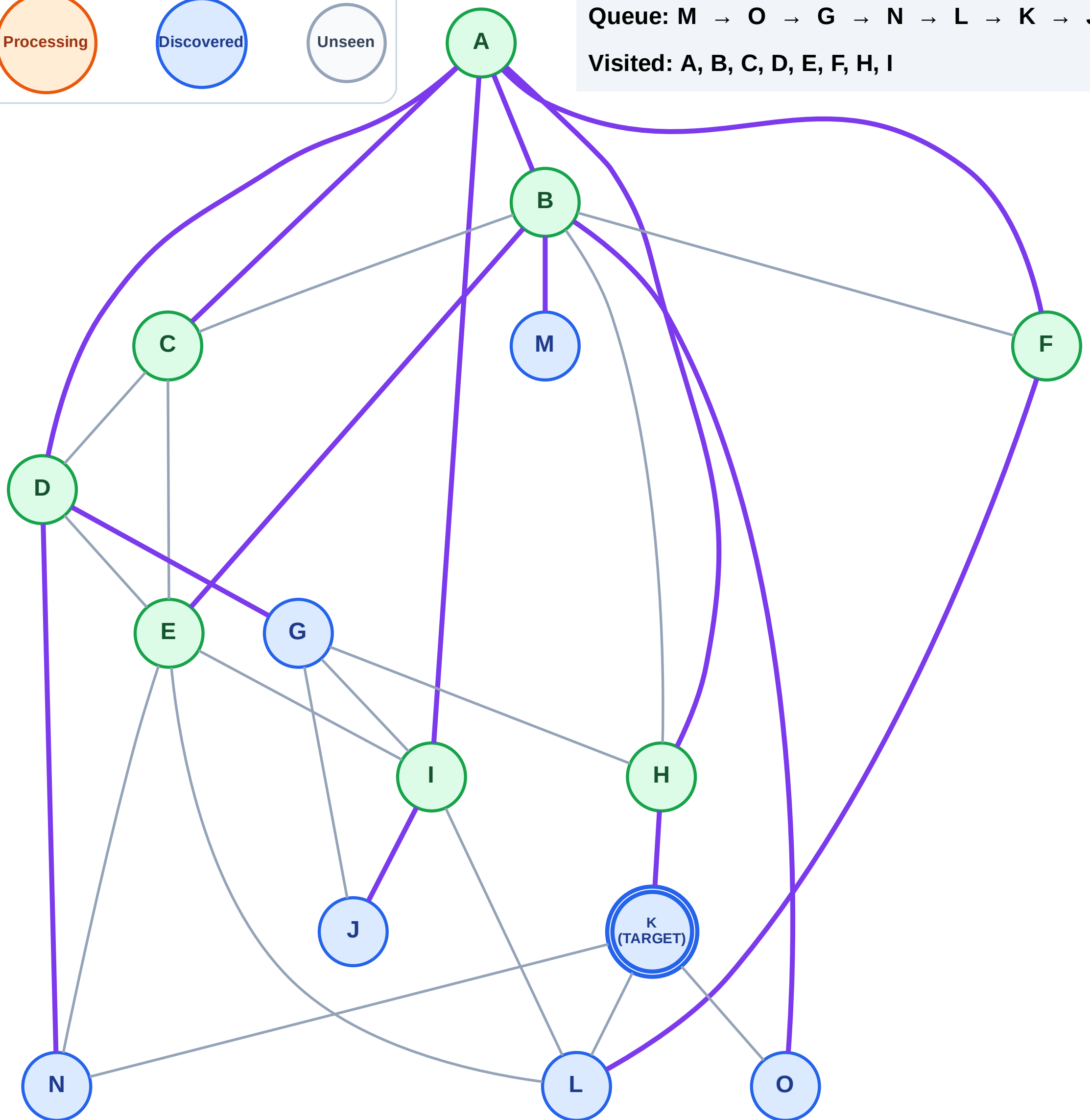
Step 17: No new neighbors from E

Legend

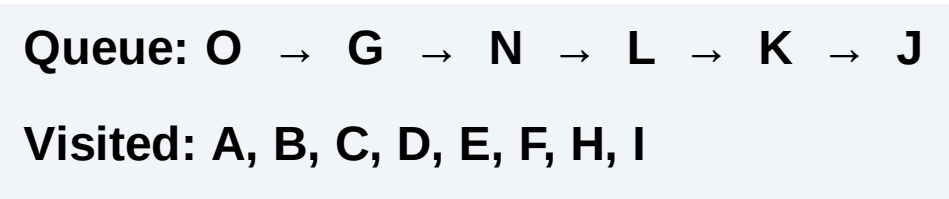


Queue: M → O → G → N → L → K → J

Visited: A, B, C, D, E, F, H, I

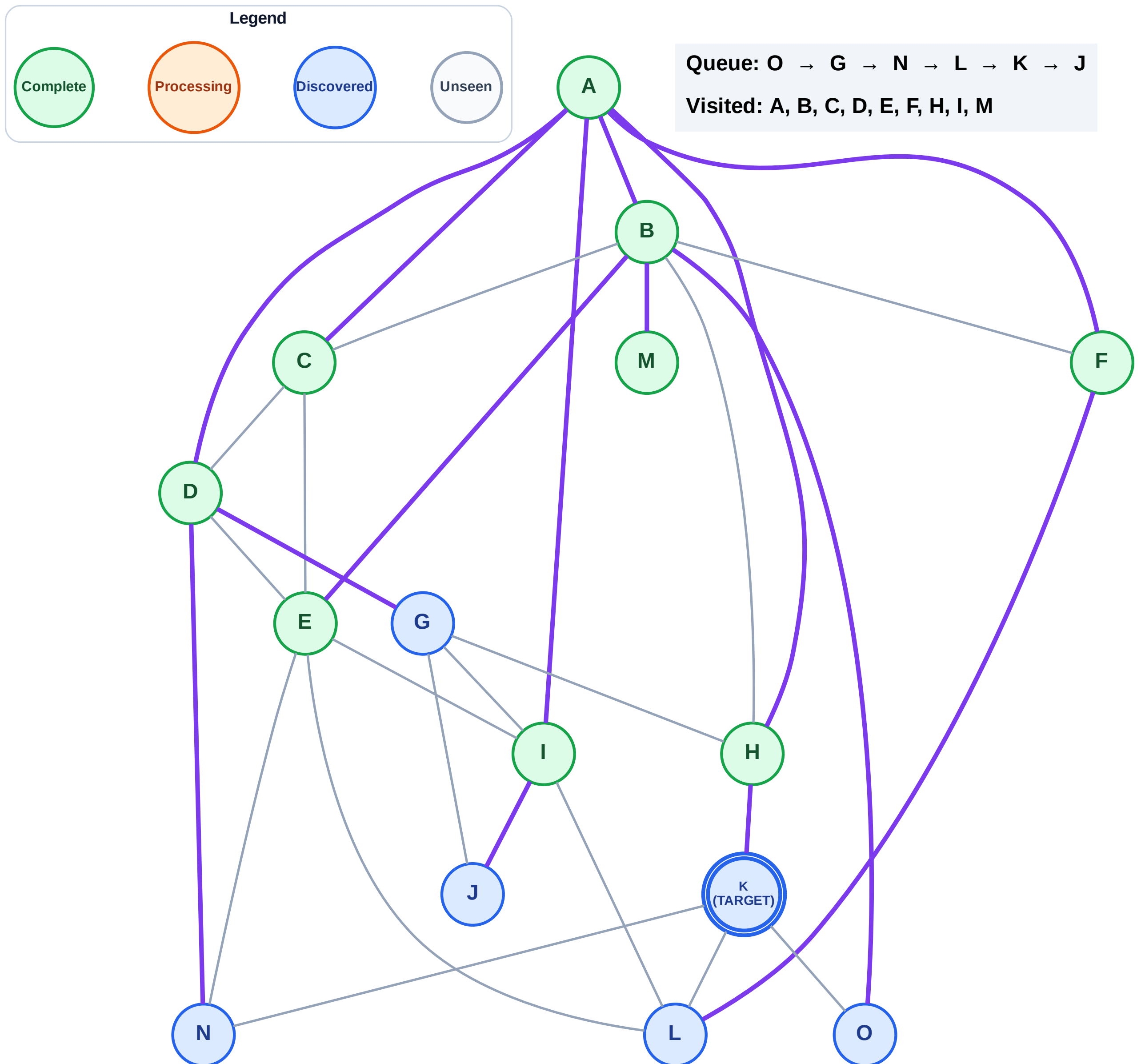


Step 18: Process node M



Step 19: No new neighbors from M

Step 19: No new neighbors from M



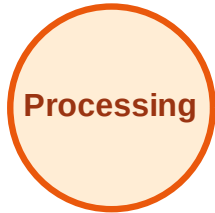
BFS Traversal

Step 20: Process node O

Legend



Complete



Processing



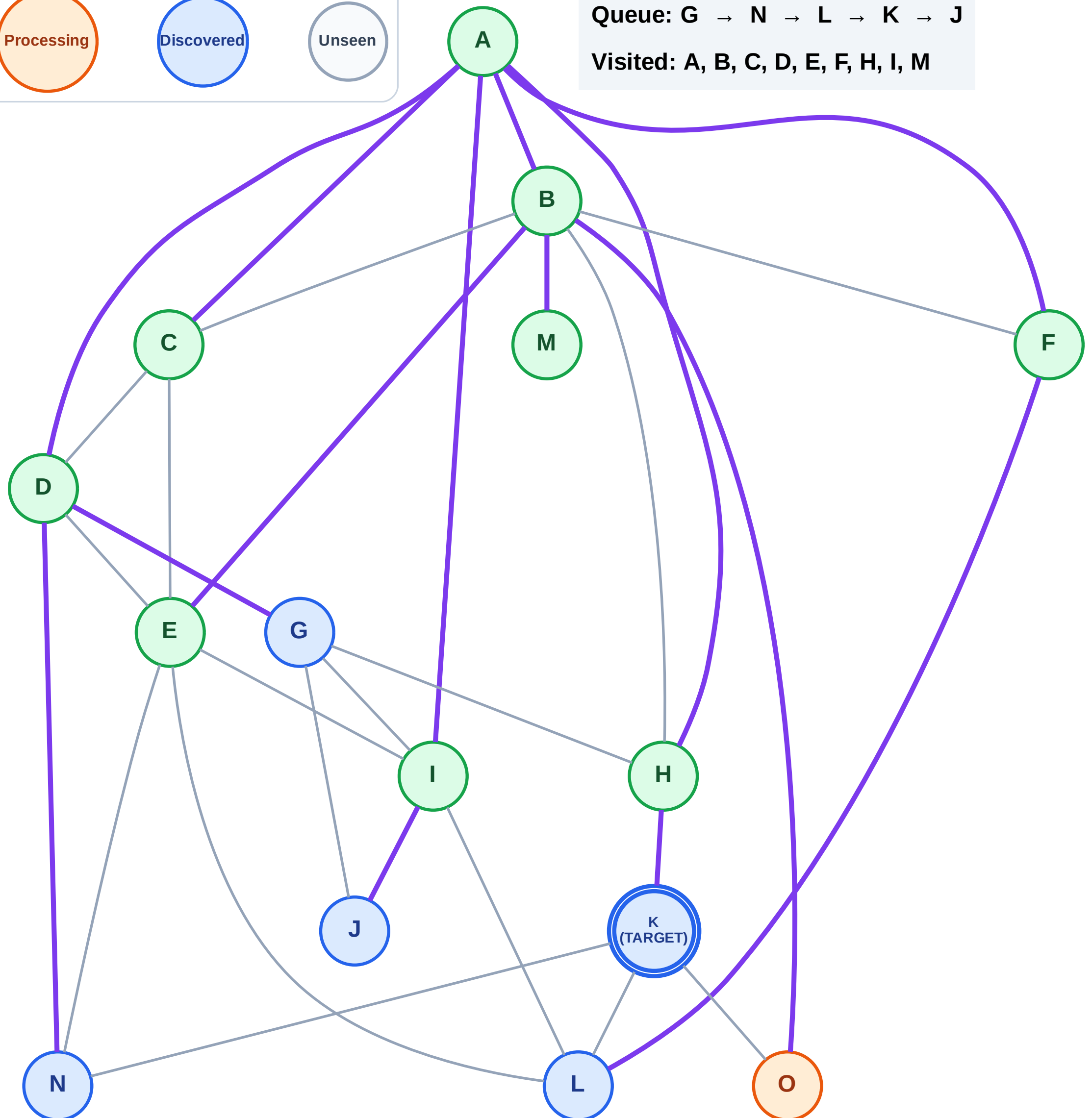
Discovered



Unseen

Queue: G → N → L → K → J

Visited: A, B, C, D, E, F, H, I, M



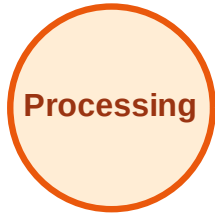
BFS Traversal

Step 21: No new neighbors from O

Legend



Complete



Processing



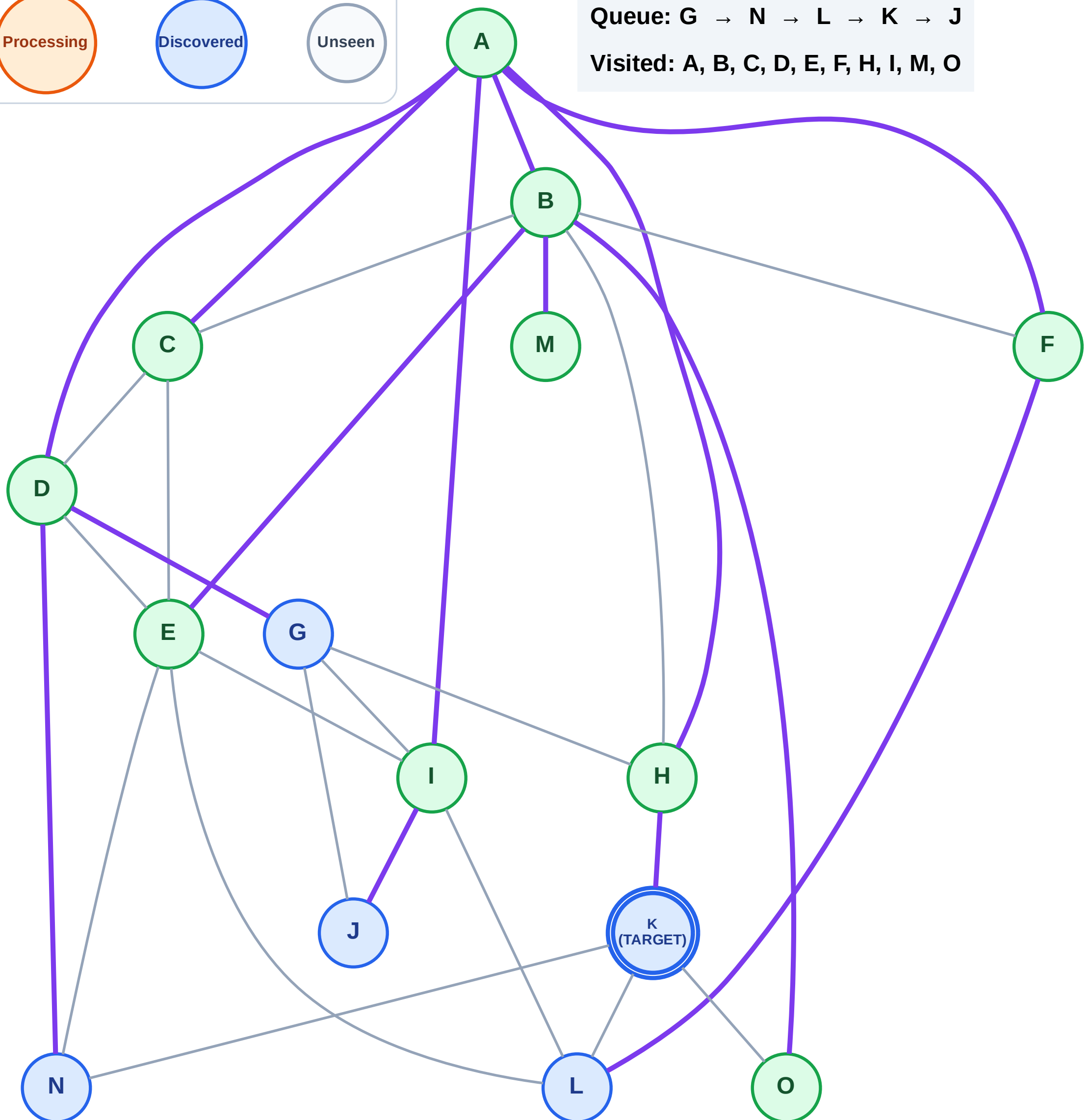
Discovered



Unseen

Queue: G → N → L → K → J

Visited: A, B, C, D, E, F, H, I, M, O



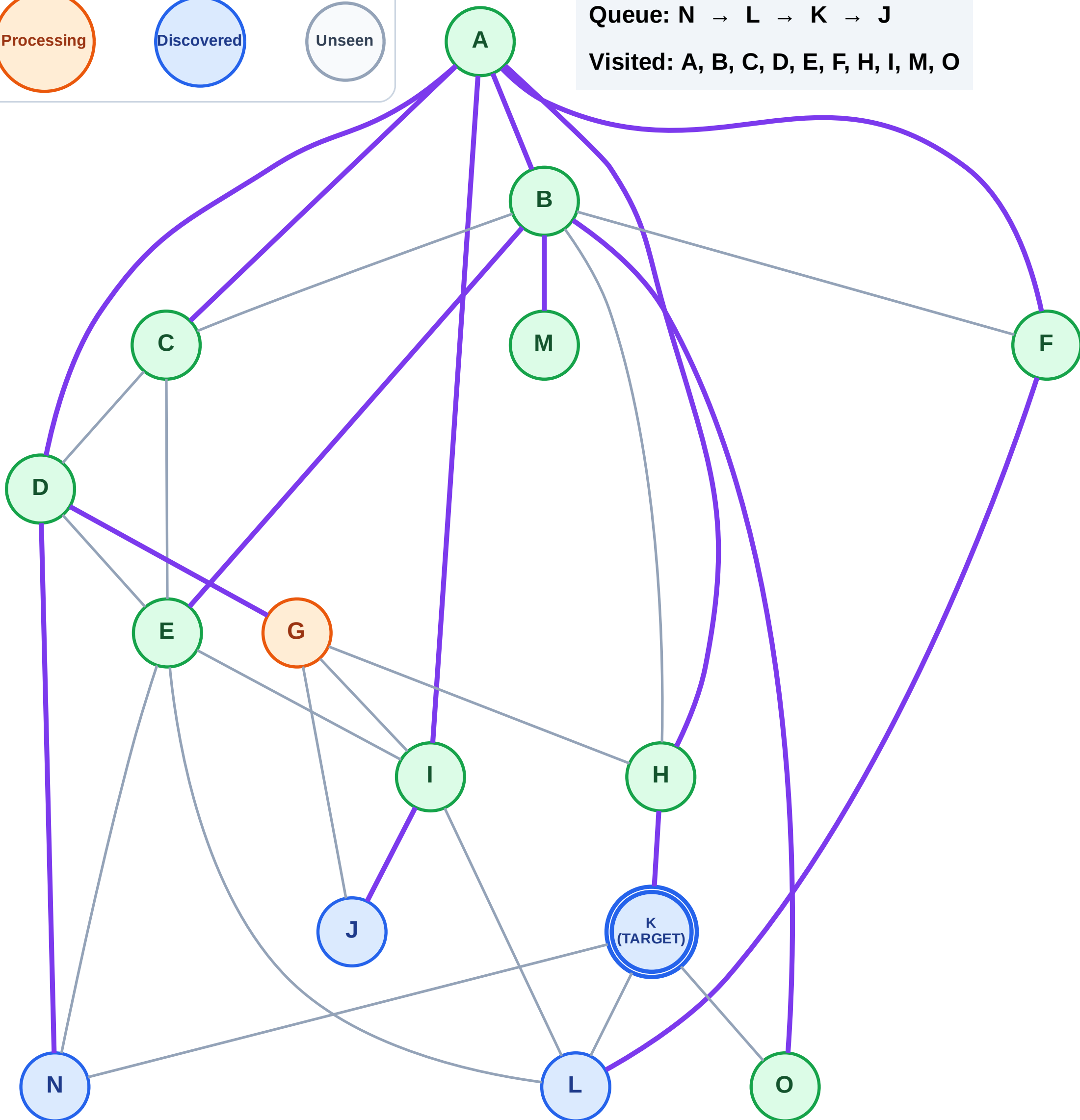
Step 22: Process node G

Legend

Processing

Unseen

Visited: A, B, C, D, E, F, H, I, M, O



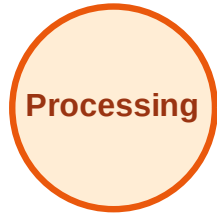
BFS Traversal

Step 23: No new neighbors from G

Legend



Complete



Processing



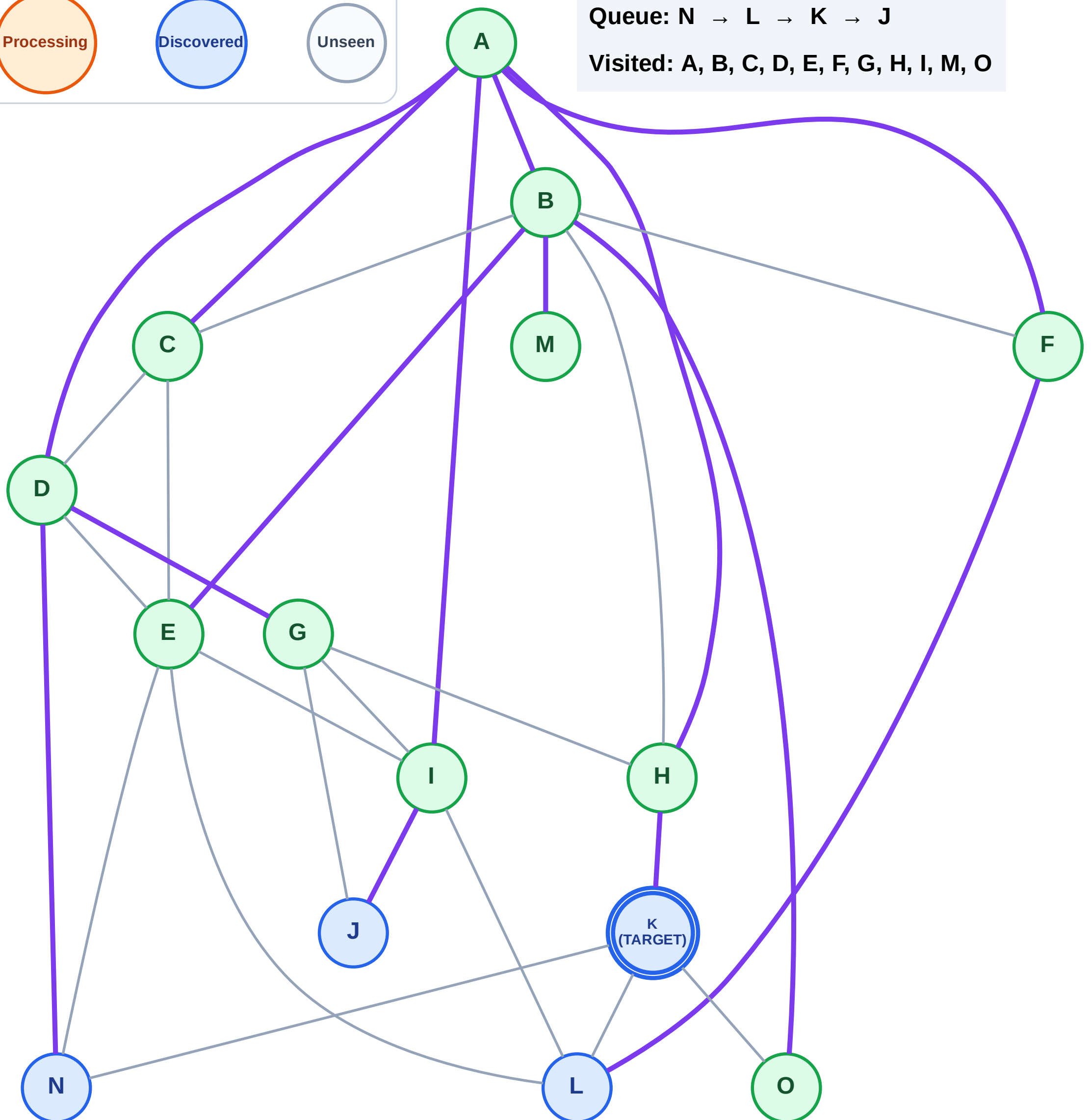
Discovered



Unseen

Queue: N → L → K → J

Visited: A, B, C, D, E, F, G, H, I, M, O



BFS Traversal

Step 24: Process node N

Legend

Complete

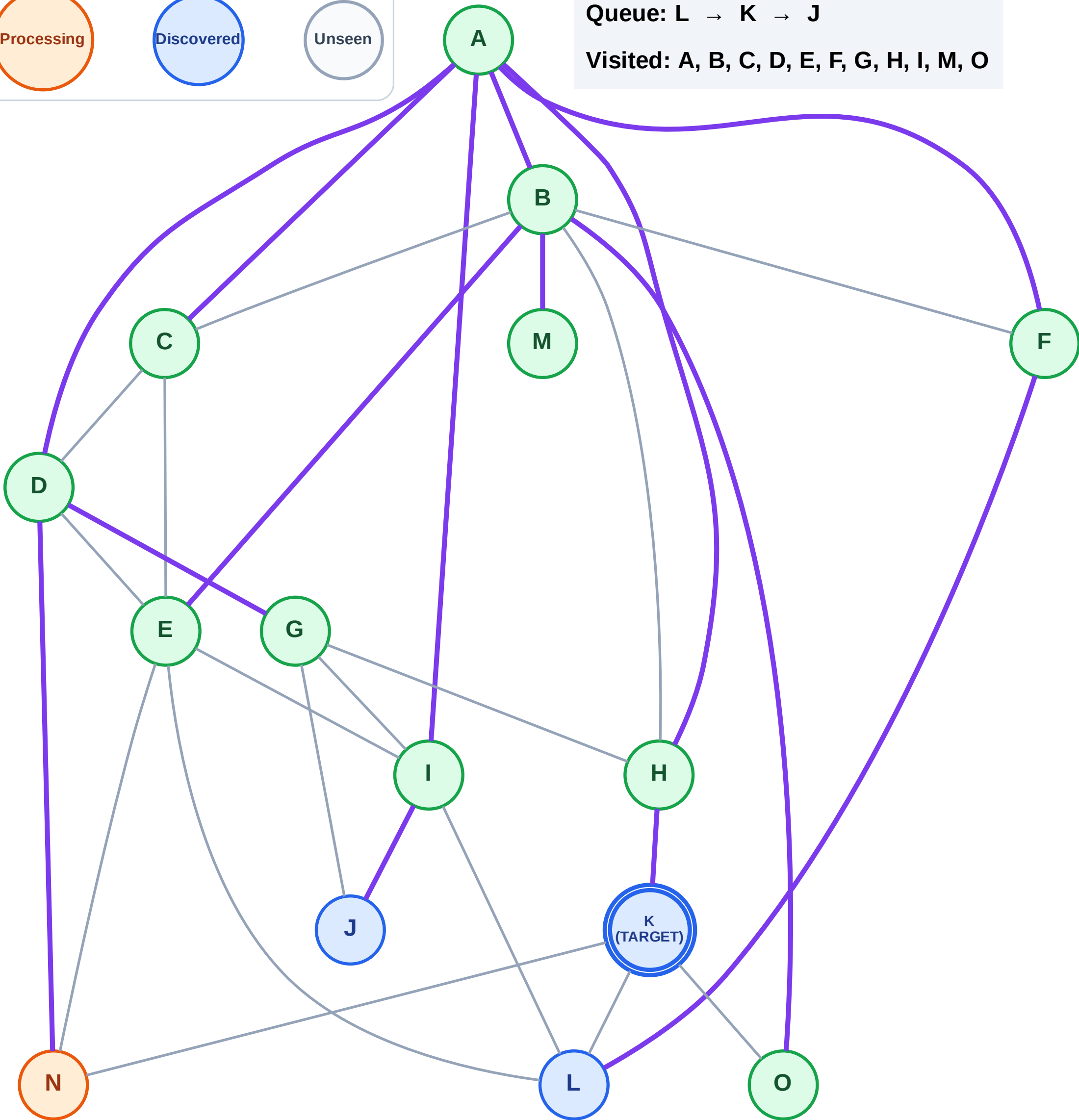
Processing

Discovered

Unseen

Queue: L → K → J

Visited: A, B, C, D, E, F, G, H, I, M, O



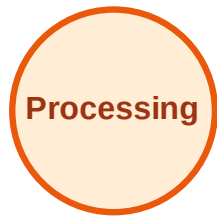
BFS Traversal

Step 25: No new neighbors from N

Legend



Complete



Processing



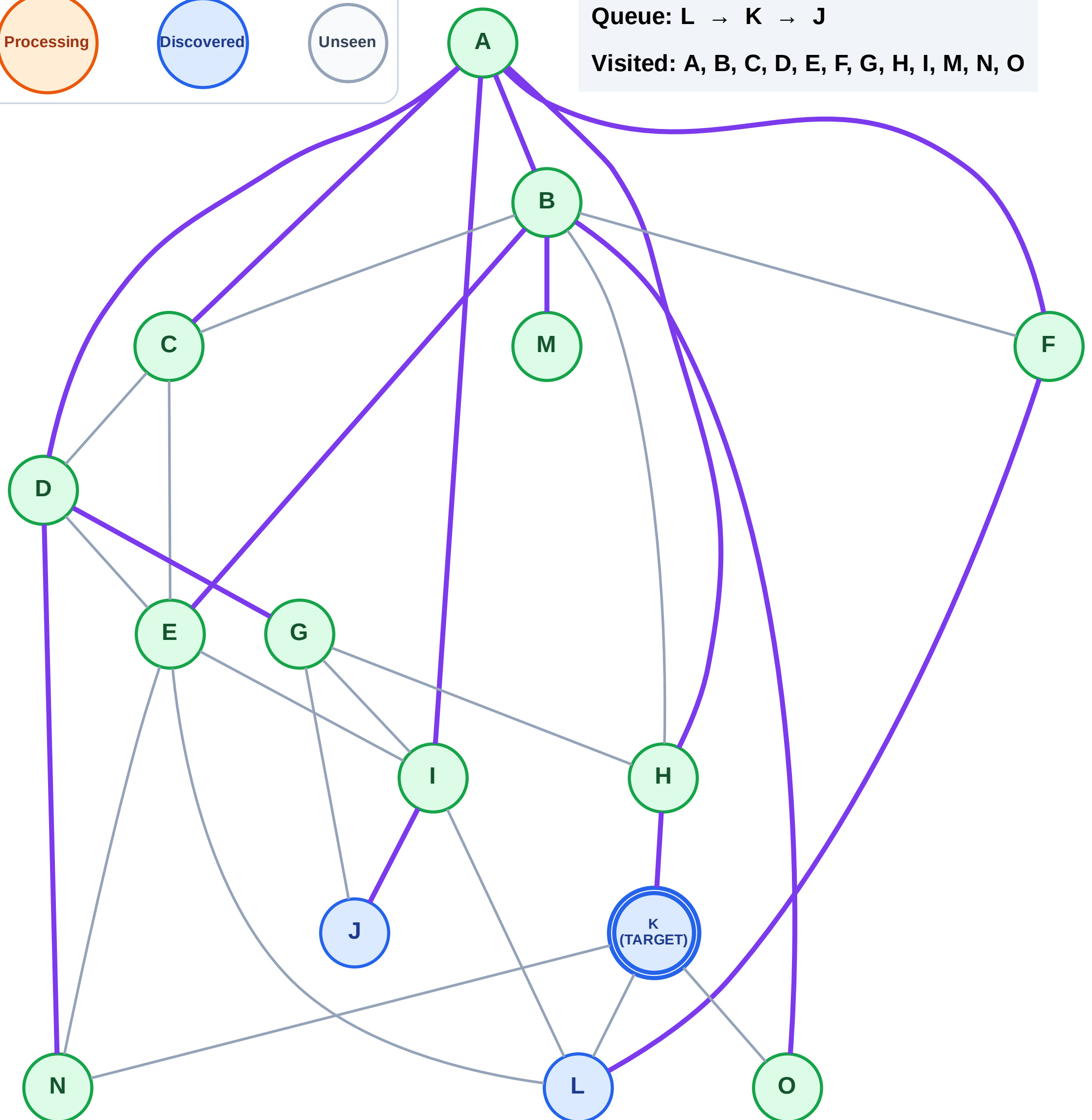
Discovered



Unseen

Queue: L → K → J

Visited: A, B, C, D, E, F, G, H, I, M, N, O



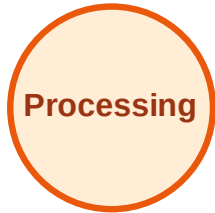
BFS Traversal

Step 26: Process node L

Legend



Complete



Processing



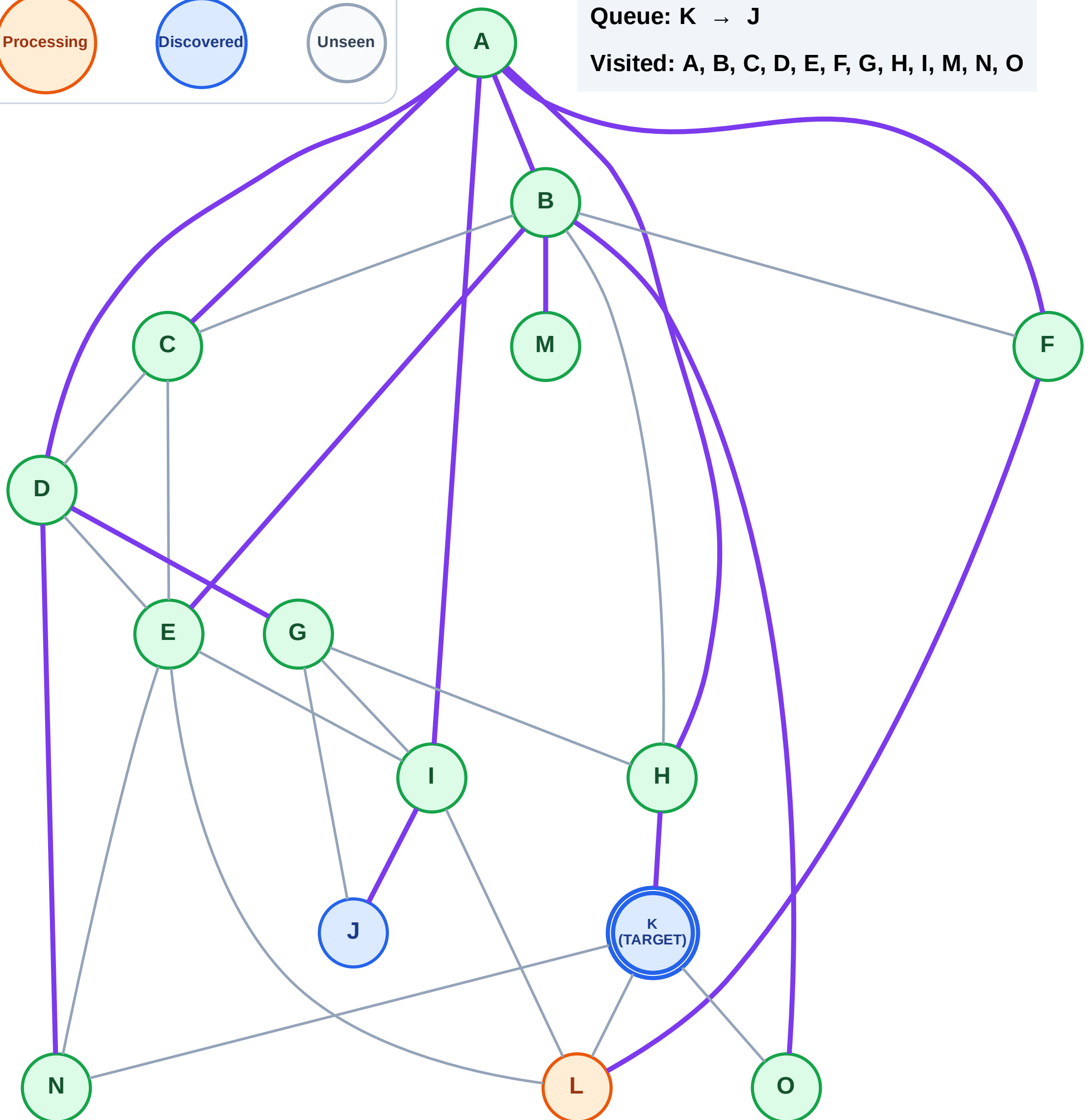
Discovered



Unseen

Queue: K → J

Visited: A, B, C, D, E, F, G, H, I, M, N, O



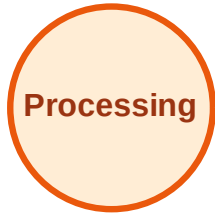
BFS Traversal

Step 27: No new neighbors from L

Legend



Complete



Processing



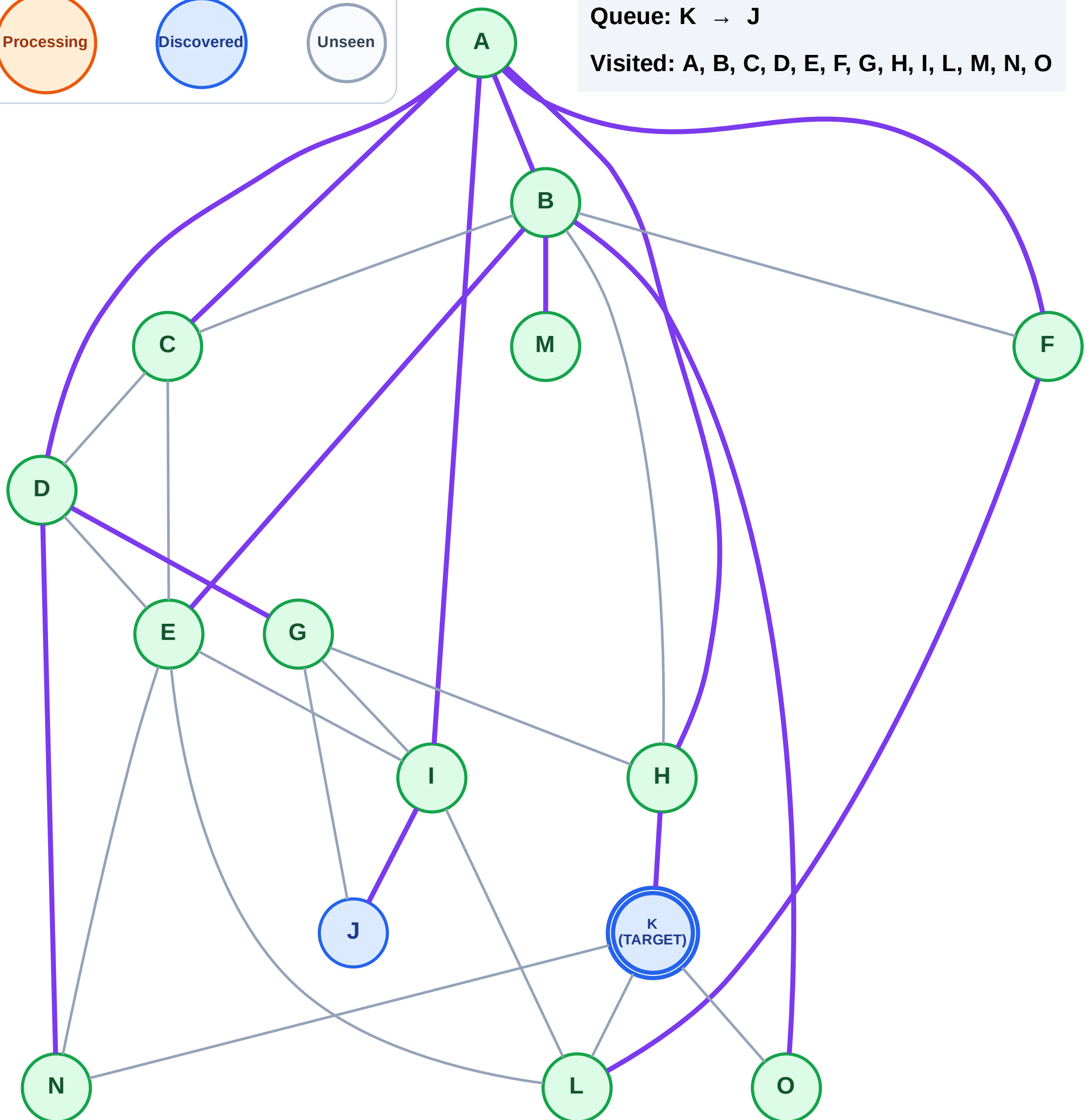
Discovered



Unseen

Queue: K → J

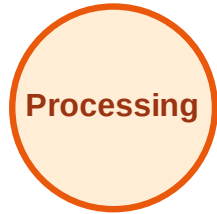
Visited: A, B, C, D, E, F, G, H, I, L, M, N, O



BFS Traversal

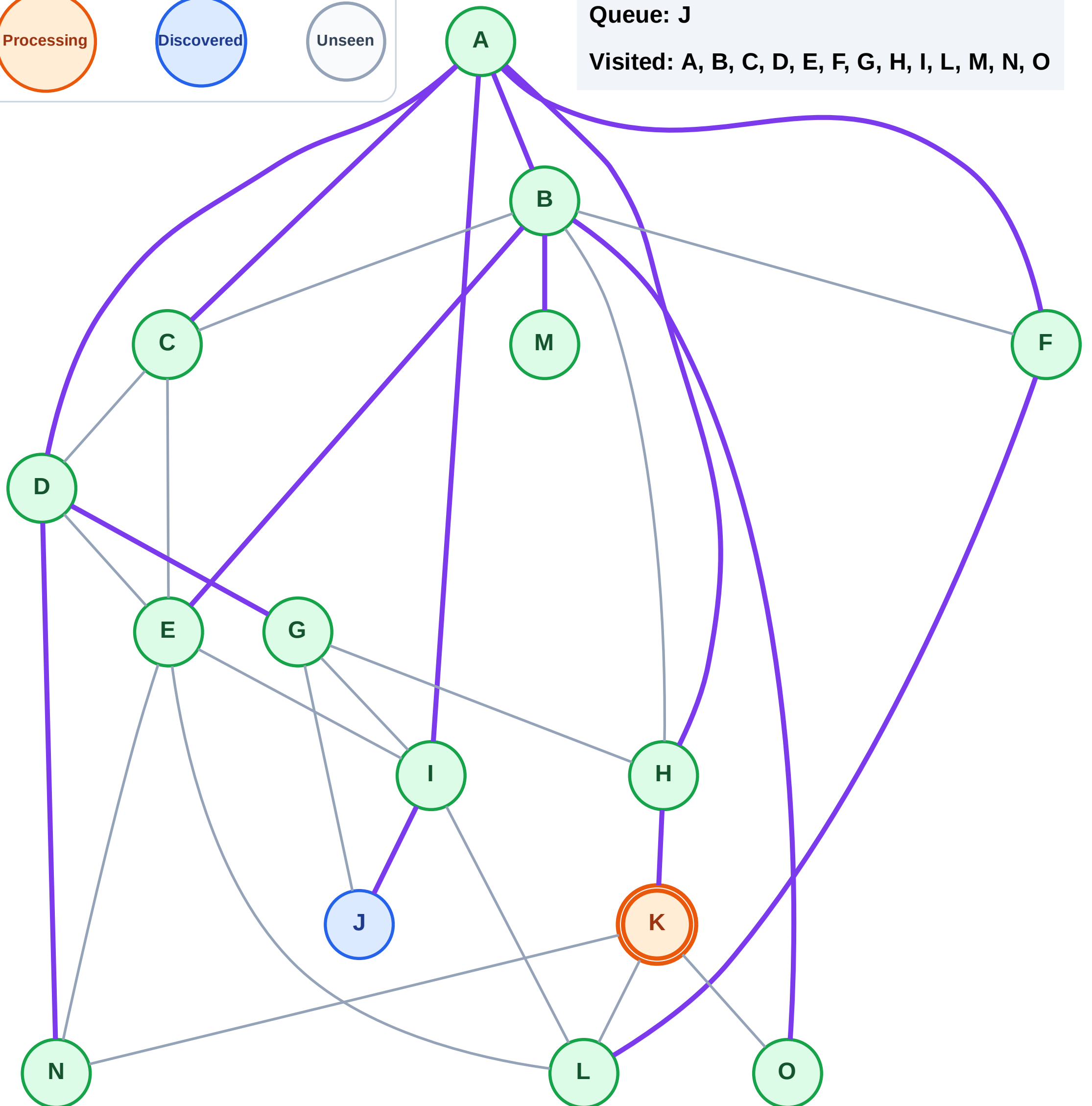
Step 28: Process node K

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I, L, M, N, O



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → H → K

Legend

Complete

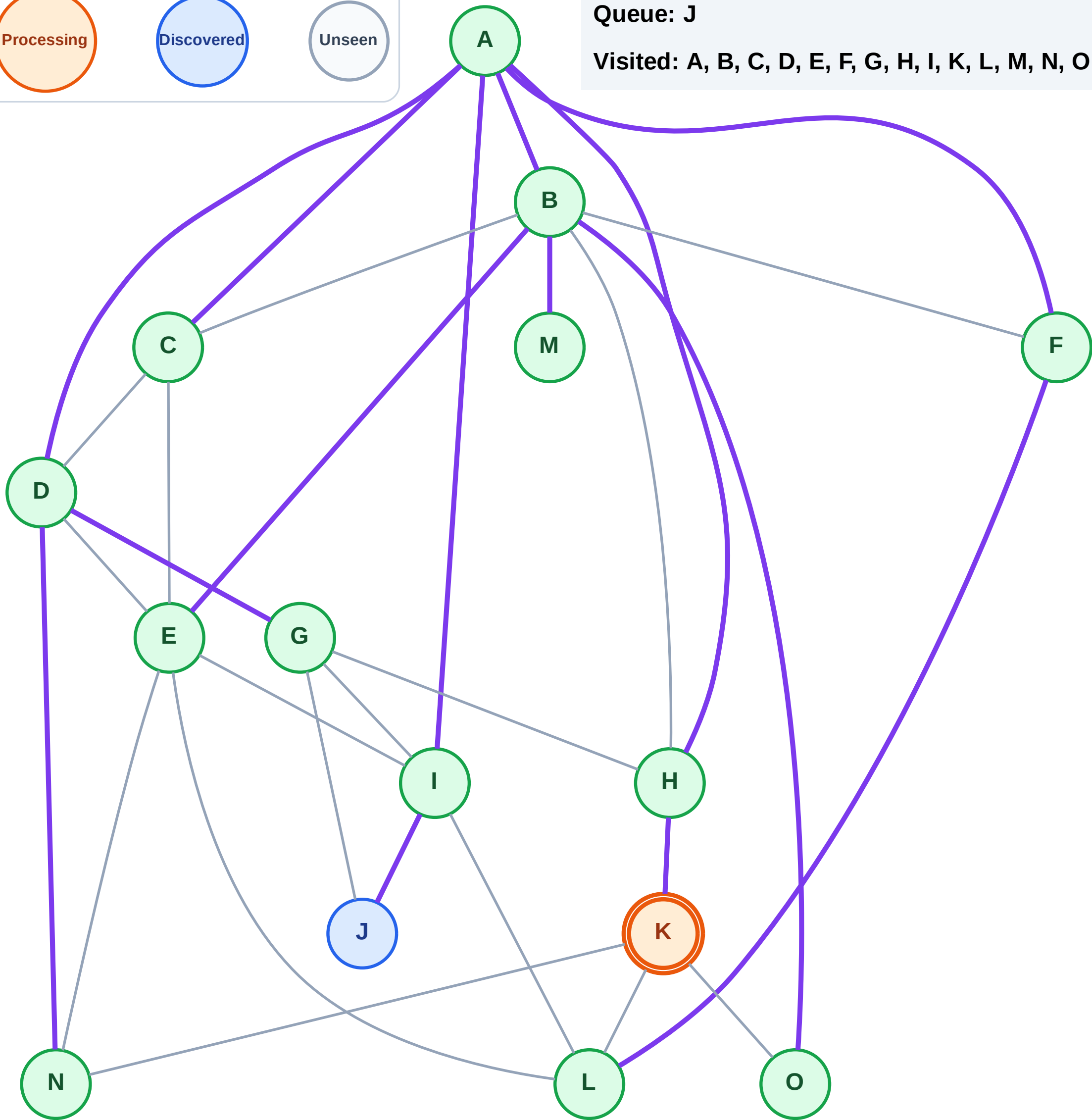
Processing

Discovered

Unseen

Queue: J

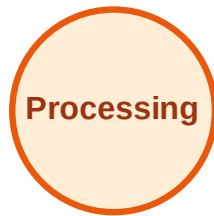
Visited: A, B, C, D, E, F, G, H, I, K, L, M, N, O



DFS Traversal

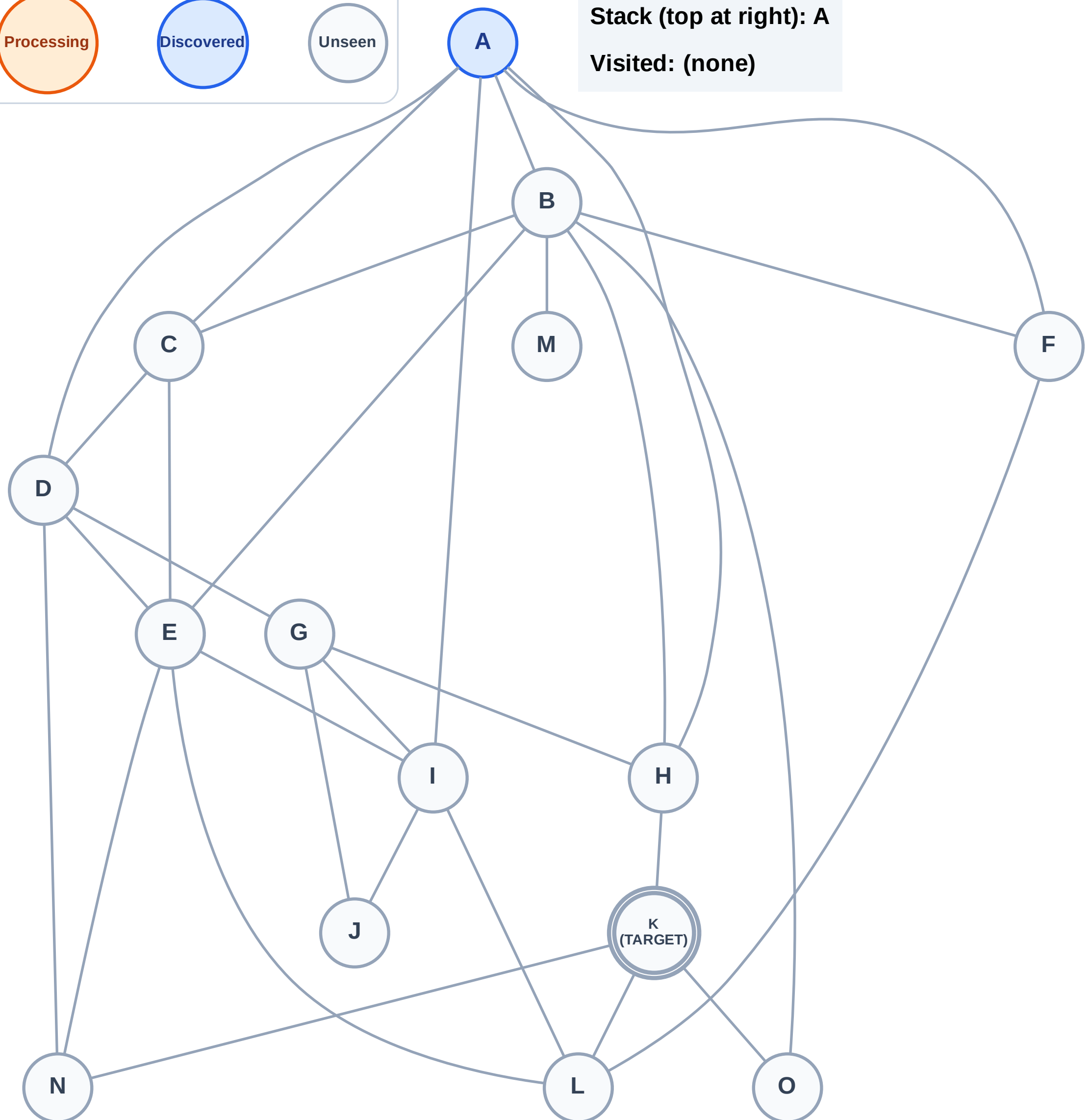
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

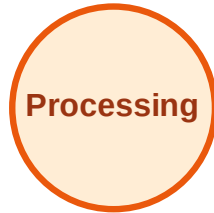
Visited: (none)



DFS Traversal

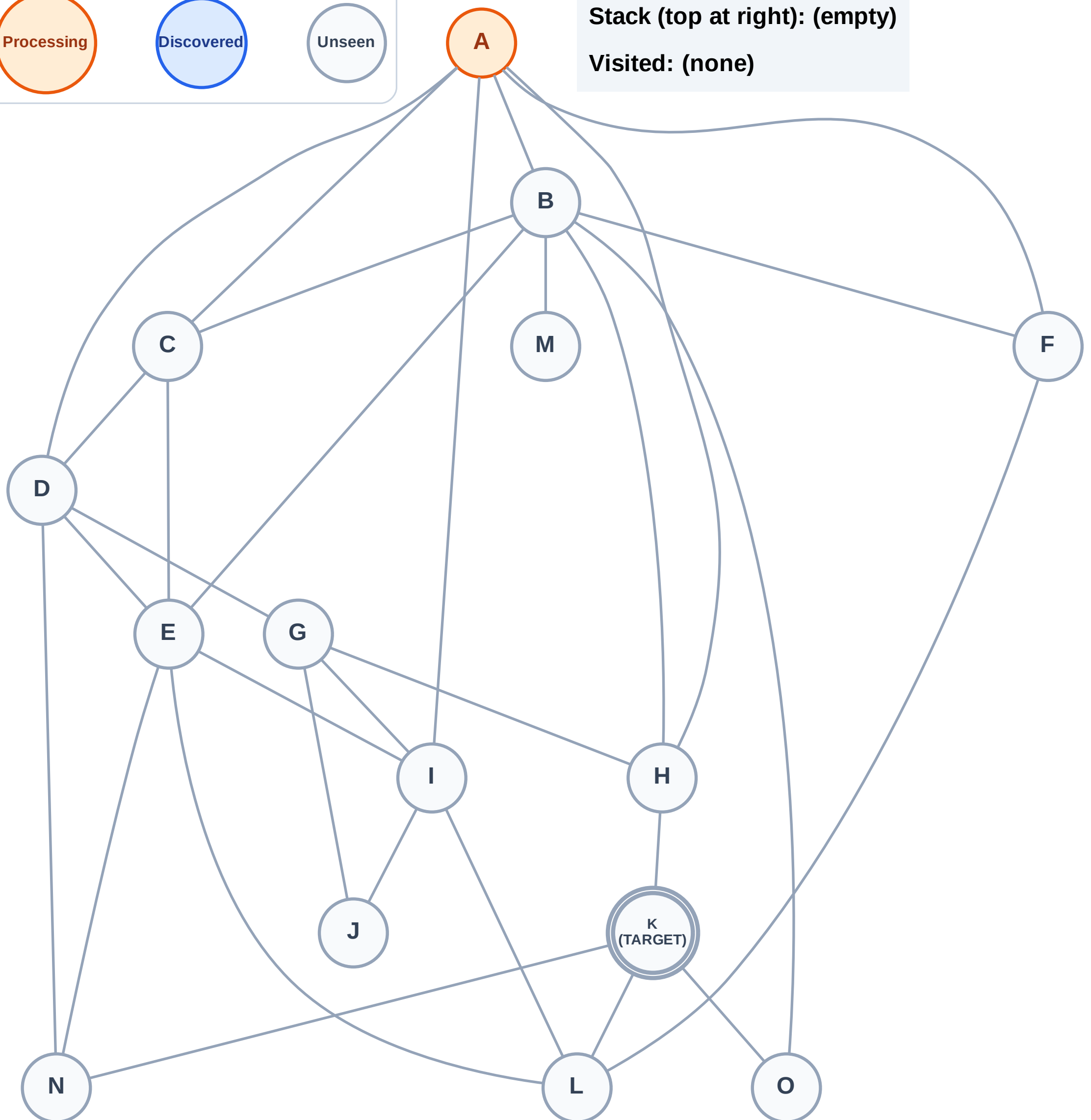
Step 2: Process node A

Legend



Stack (top at right): (empty)

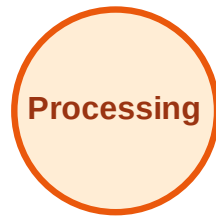
Visited: (none)



DFS Traversal

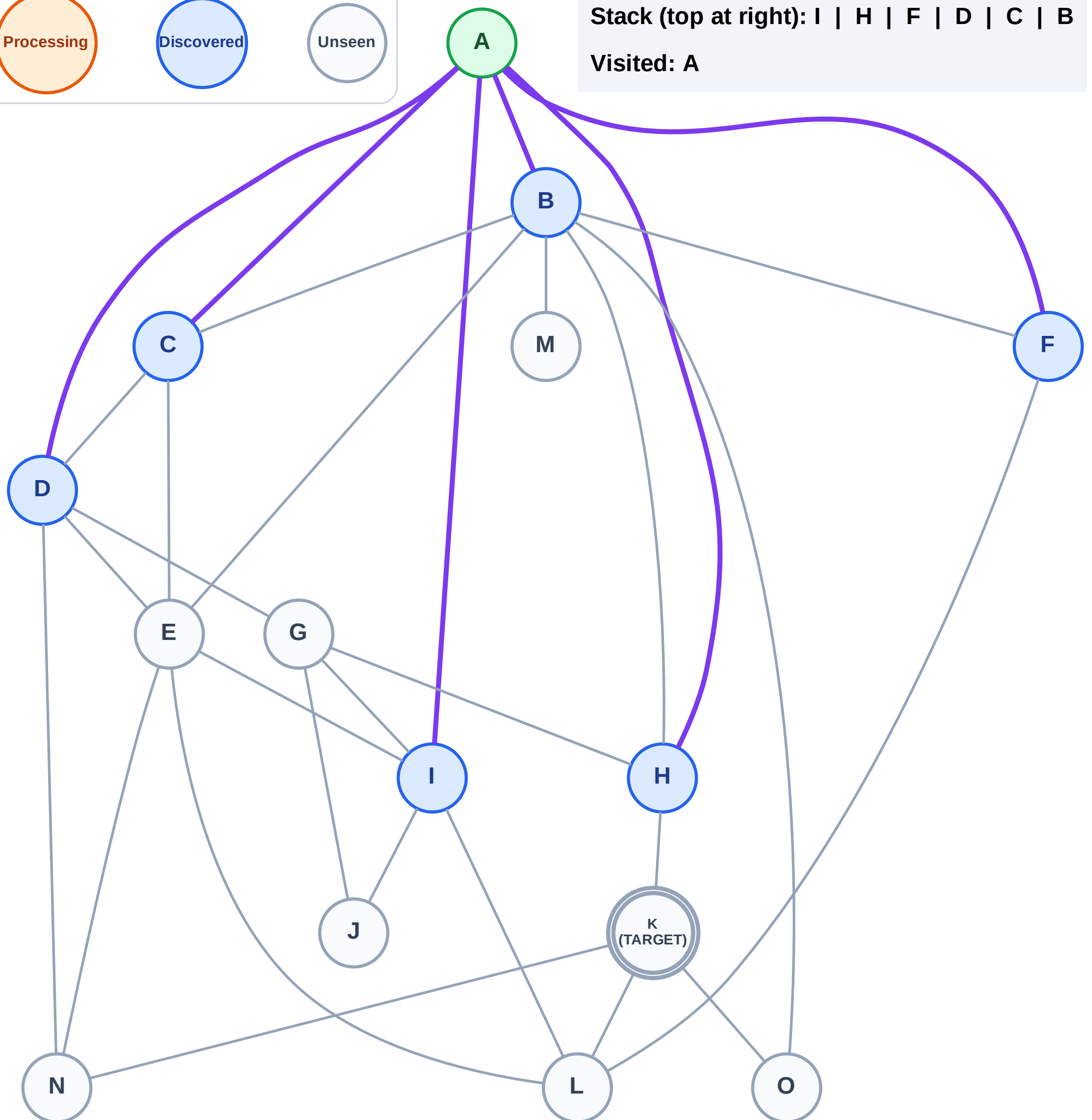
Step 3: Discover I, H, F, D, C, B from A

Legend



Stack (top at right): I | H | F | D | C | B

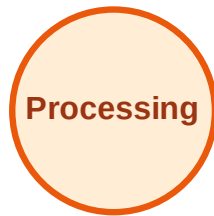
Visited: A



DFS Traversal

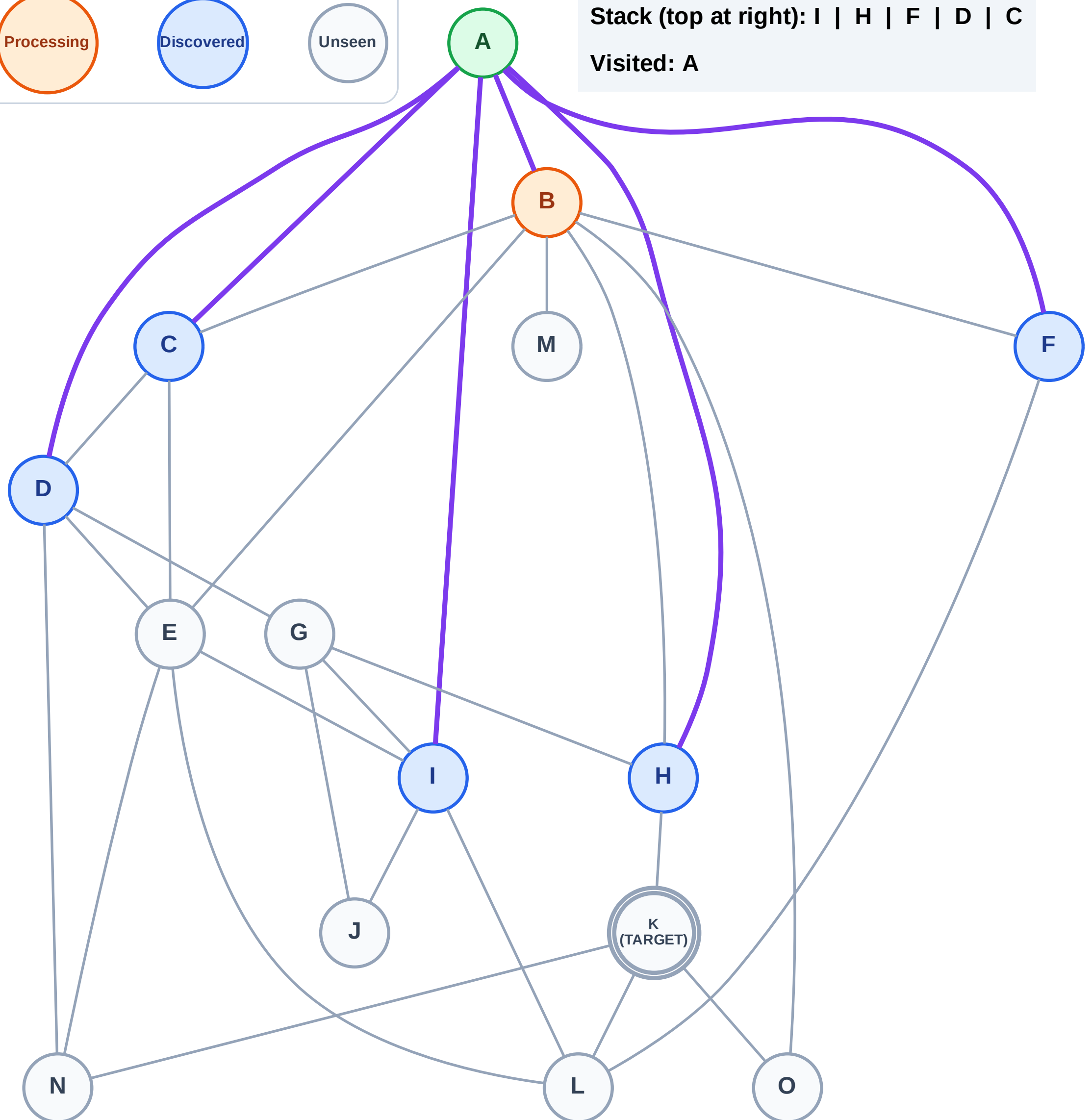
Step 4: Process node B

Legend



Stack (top at right): I | H | F | D | C

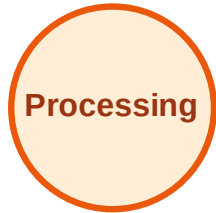
Visited: A



DFS Traversal

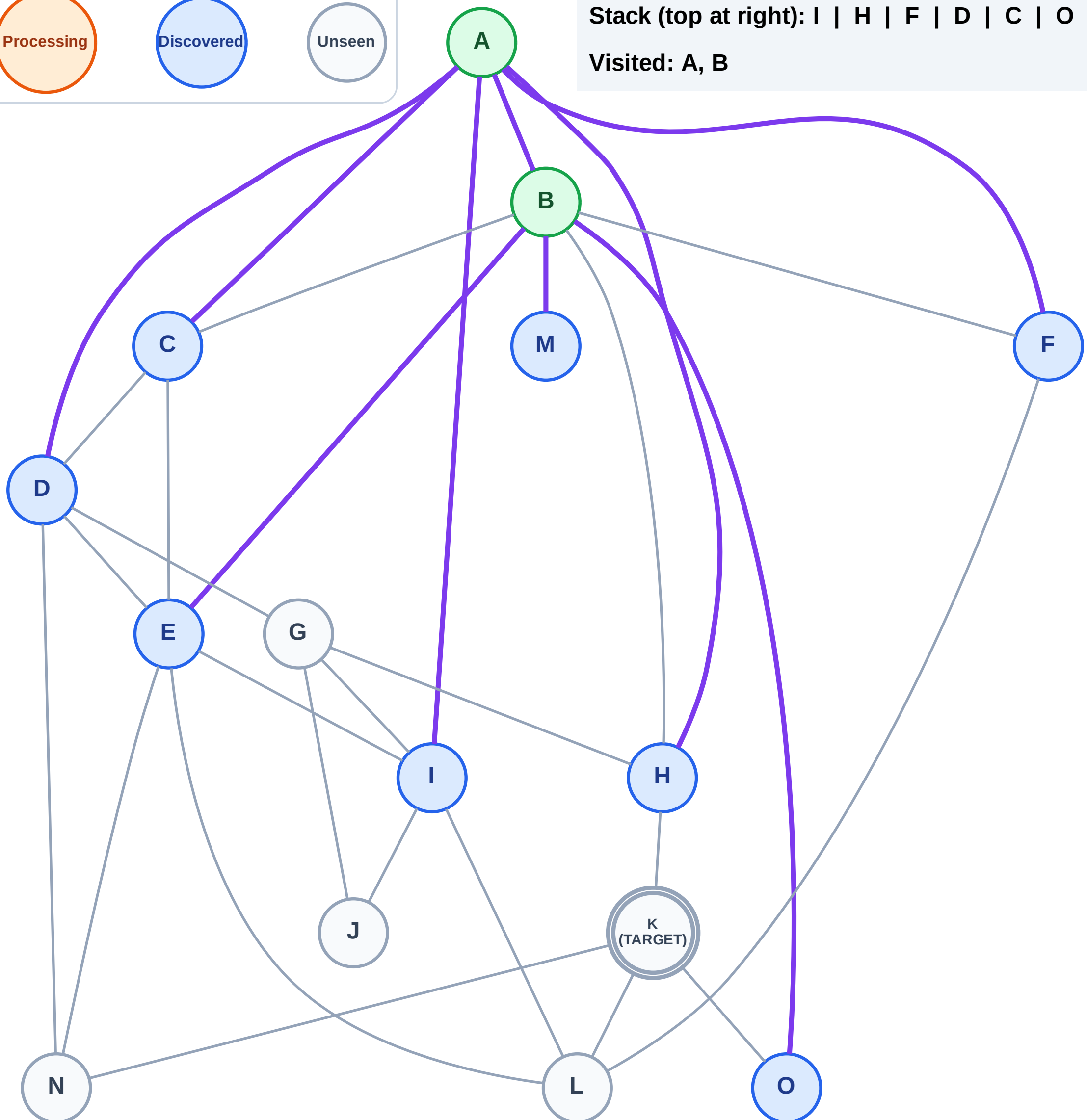
Step 5: Discover O, M, E from B

Legend



Stack (top at right): I | H | F | D | C | O | M | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

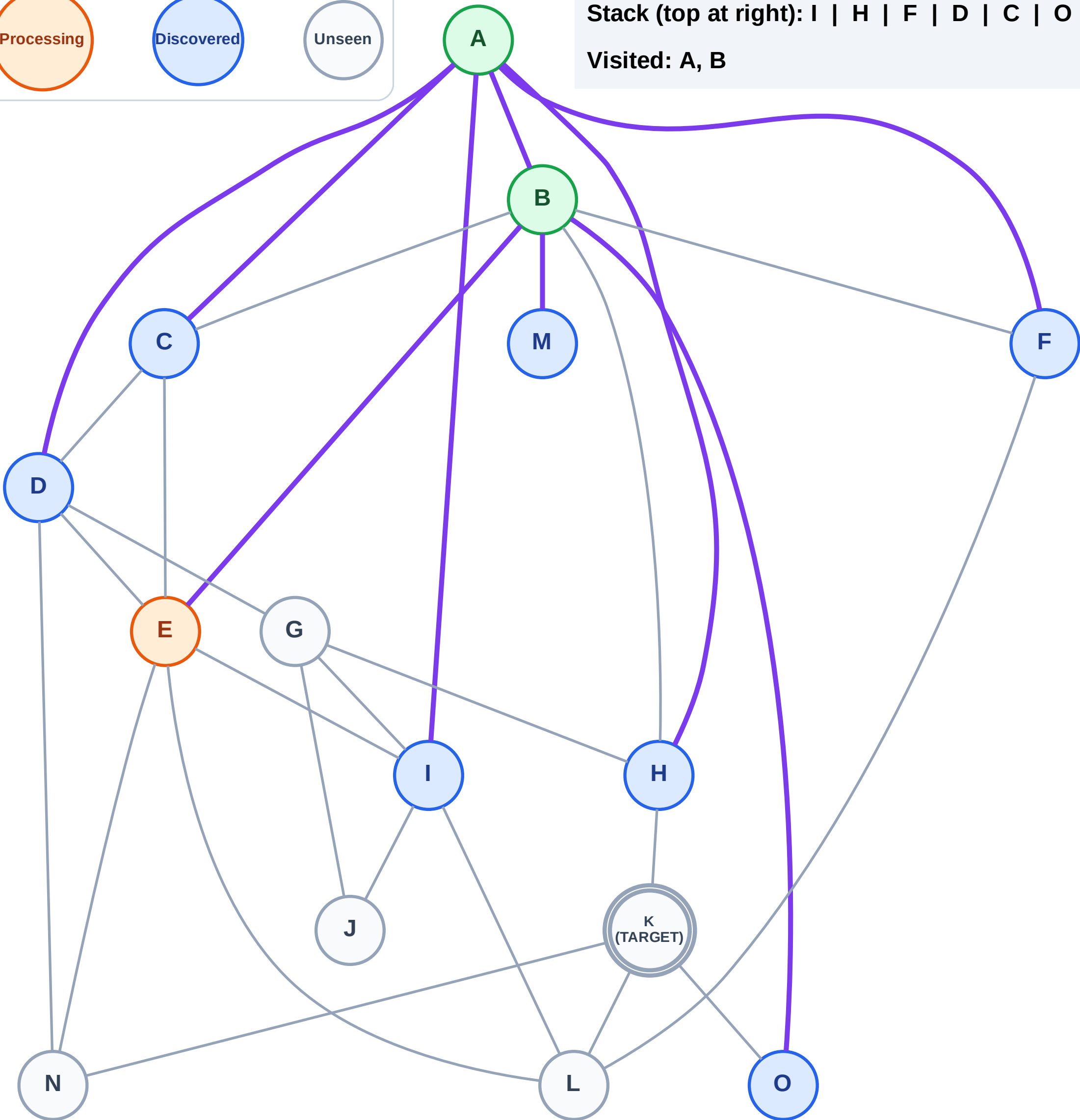
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | C | O | M

Visited: A, B



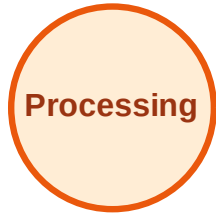
DFS Traversal

Step 7: Discover N, L from E

Legend



Complete



Processing



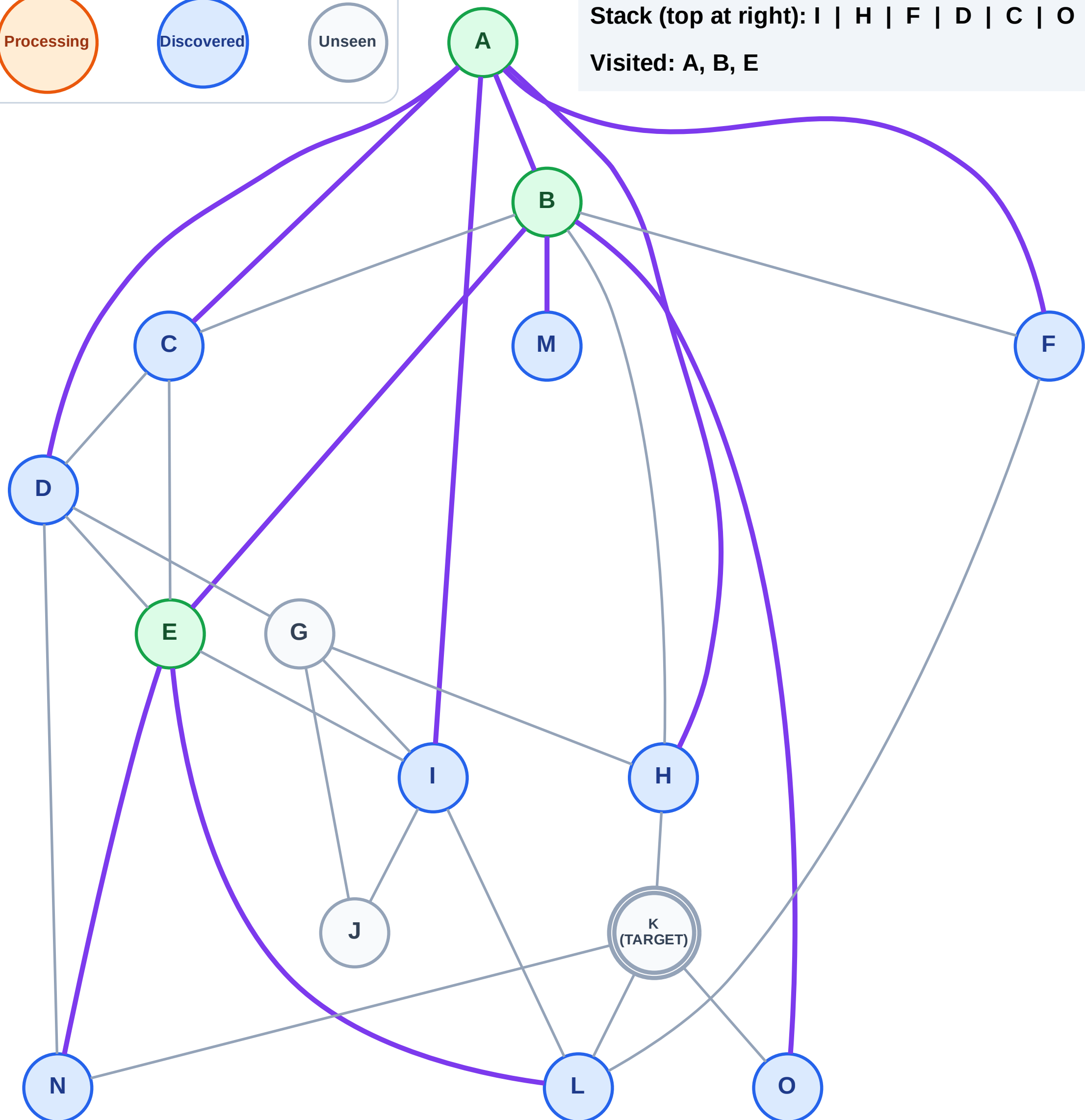
Discovered



Unseen

Stack (top at right): I | H | F | D | C | O | M | N | L

Visited: A, B, E



DFS Traversal

Step 8: Process node L

Legend

Complete

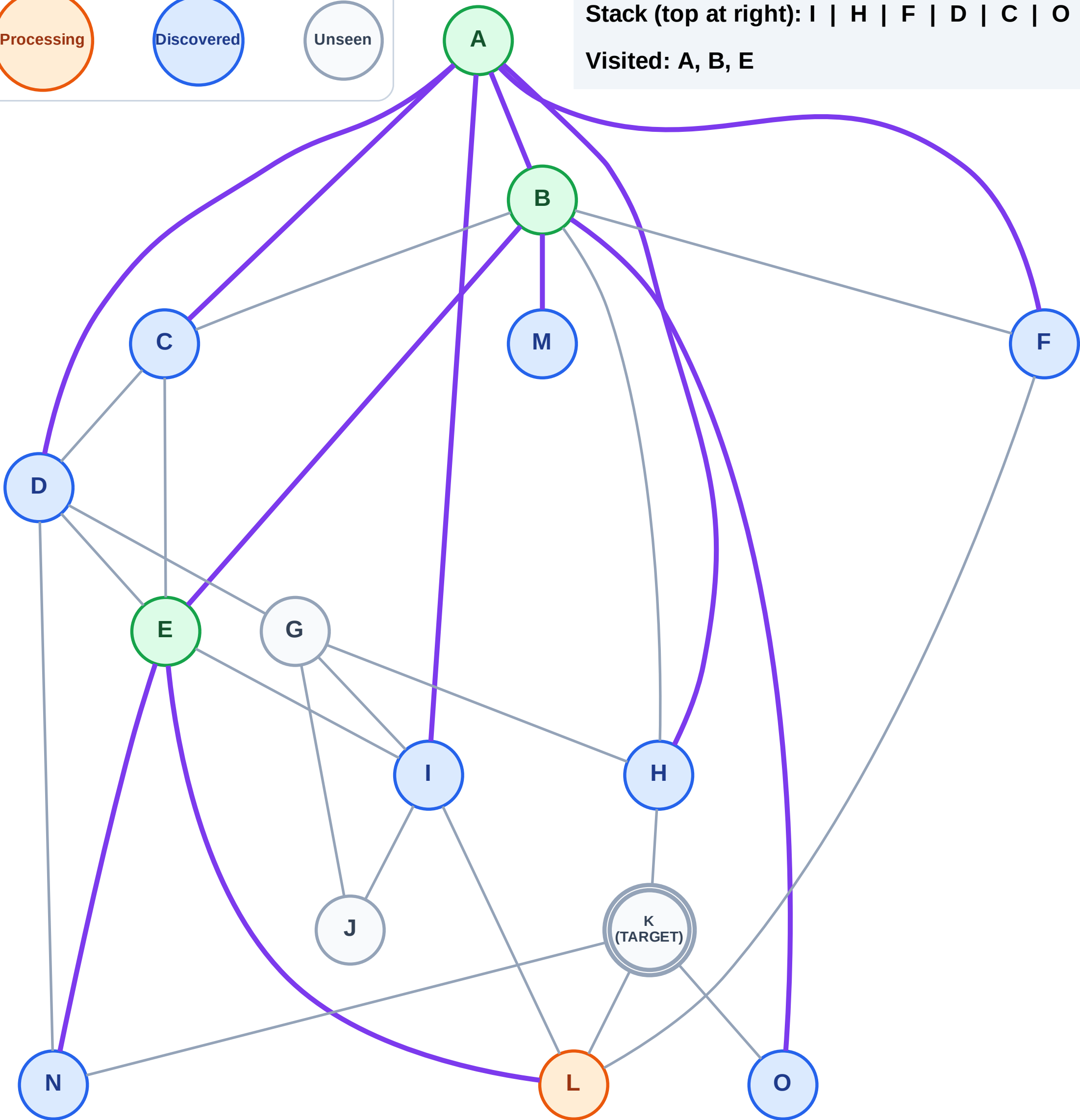
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | C | O | M | N

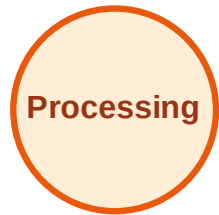
Visited: A, B, E



DFS Traversal

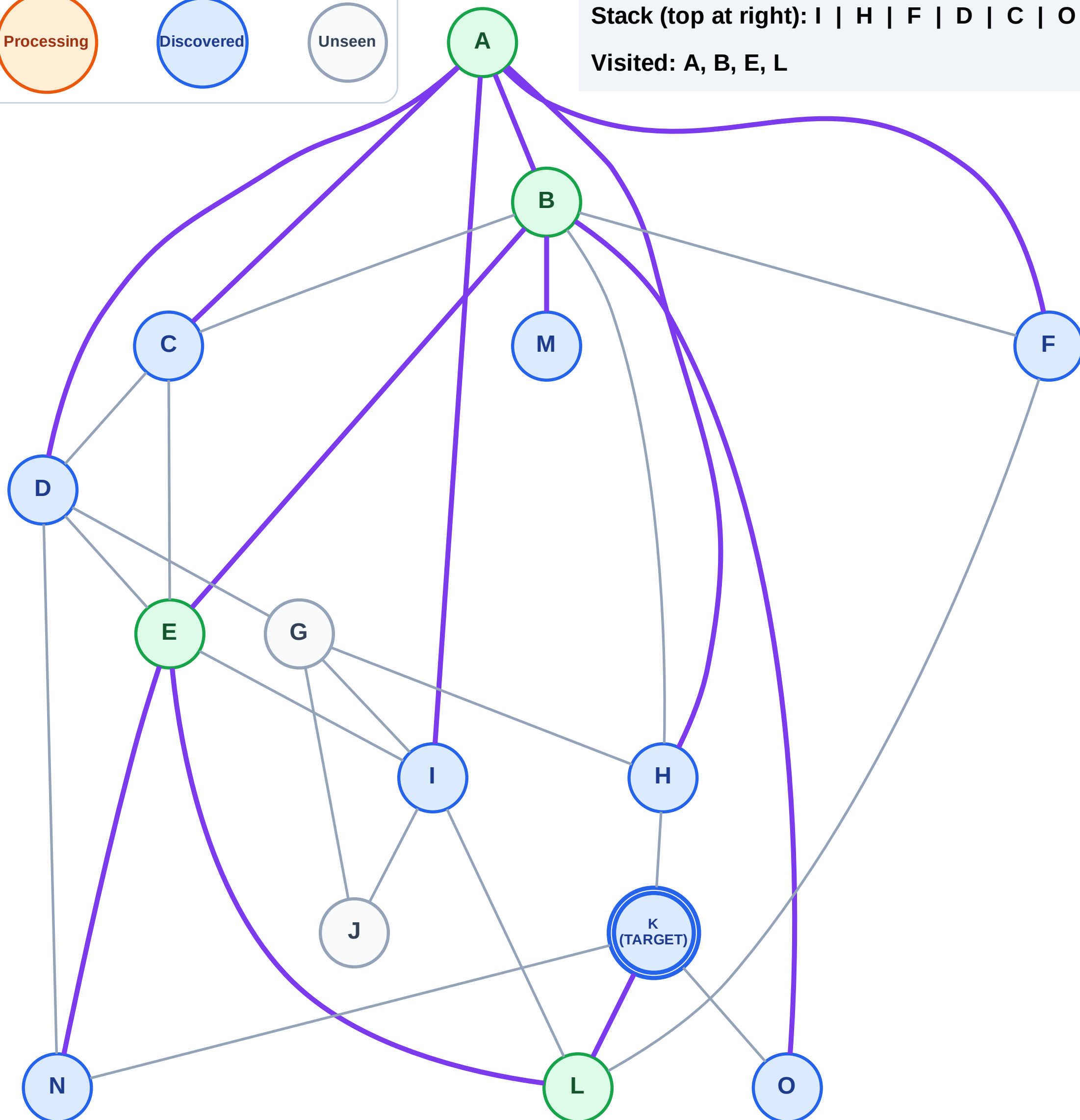
Step 9: Discover K from L

Legend

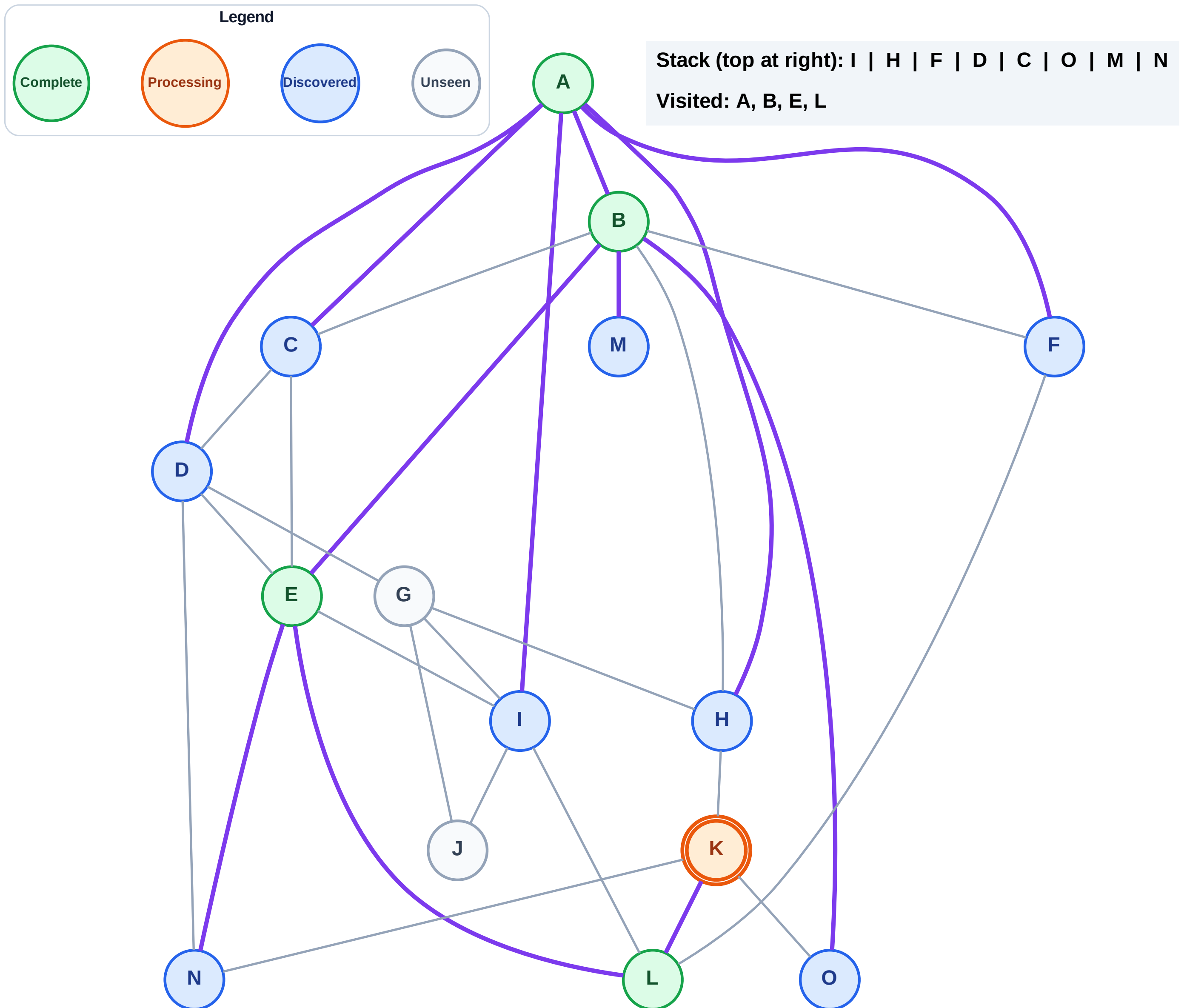


Stack (top at right): I | H | F | D | C | O | M | N | K

Visited: A, B, E, L



Step 10: Process node K



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (4 hops): A → B → E → L → K

Legend



Stack (top at right): I | H | F | D | C | O | M | N

Visited: A, B, E, K, L

